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Reciprocal Relationships between Parental Burnout and Early Adolescent Loneliness: An Analysis of Parent-adolescent Gender Dyads

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Abstract: To examine the reciprocal relationships between parental burnout and early adolescent loneliness and the role of parent-adolescent gender dyads, this study employed a two-wave cross-lagged design, with data collected when adolescents were in Grade 7 and again in Grade 8. Fathers' and mothers' parental burnout was assessed using the Parental Burnout Assessment, and early adolescents' loneliness was assessed using the Children and Adolescents Loneliness Scale. Data from 987 families were included in the final analysis. In Grade 7, early adolescents had a mean age of 13.54 ± 0.34 years, fathers had a mean age of 43.34 ± 5.59 years, and mothers had a mean age of 41.31 ± 4.63 years. After controlling for subjective family socioeconomic status, cross-lagged panel analyses revealed that, among boys, loneliness positively predicted both fathers' and mothers' parental burnout. Among girls, mothers' parental burnout positively predicted girls' loneliness. Furthermore, no reciprocal relationships were found between fathers' or mothers' parental burnout and boys' or girls' loneliness. These findings indicate that the associations between parental burnout and early adolescent loneliness are characterized by unidirectional predictive patterns that vary across parent-adolescent gender dyads.

Key words: parental burnout; loneliness; reciprocal relationships; parent-adolescent gender dyads; early adolescents

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1 Introduction

Along with sustained socioeconomic development, Chinese parents have increasingly valued scientific child-rearing and high-quality companionship for their children. Higher parenting expectations, however, require parents to invest greater time and emotional resources. When parenting demands and parenting resources are imbalanced over the long term, parents are prone to parental burnout. Existing research indicates that parental burnout not only undermines parents' own mental health but may also give rise to adolescent maladjustment (Chen et al., 2022; Mikolajczak et al., 2018; Yang et al., 2021; Bian et al., 2026; Li et al., 2024; Ping et al., 2022; Wang et al., 2021). Adolescence is a peak period of the need for emotional connection and a sensitive period for the experience of loneliness (Qualter et al., 2015). The Chinese National Mental Health Development Report (2021–2022) shows that about 40% of adolescents report that they "often feel lonely"; particularly in early adolescence, after individuals enter junior high school, loneliness rises markedly and continues to increase from Grade 7 to Grade 8 (Fu et al., 2023). Yet burned-out parents may reduce the emotional support they provide to adolescents, intensifying early adolescents' experience of loneliness (Cheng et al., 2020). At the same time, parenting an adolescent child is in itself a highly challenging task. Adolescents are emotionally sensitive and strive for independence, and parent–child interactions decrease while conflicts increase (Laursen & Collins, 2004); these characteristics may also aggravate parents' frustration and emotional exhaustion. In other words, early adolescents' maladjustment may in turn trigger parental burnout (Hong et al., 2022; Minh et al., 2022; Ren et al., 2024).

Does parental burnout predict early adolescent loneliness, does early adolescent loneliness predict parental burnout, or do both operate? A study based on Western samples suggests that reciprocal relationships may exist between parental burnout and children's and adolescents' internalizing and externalizing problems (Woine et al., 2026). However, direct longitudinal empirical evidence is still lacking on whether reciprocal relationships exist between Chinese parents' burnout and early adolescent loneliness. Of particular note, gender affects not only role expectations within the family but also emotional expression and communication styles, so different parent–child dyads may be characterized by distinct interaction processes. This implies that the pattern of the relationship between parental burnout and early adolescent loneliness may also differ across parent-adolescent gender dyads, yet existing research offers a relatively limited understanding of this important process. In sum, drawing on two-wave longitudinal data and using a cross-lagged design, this study examines the reciprocal relationships between Chinese parents' parental burnout and early adolescent loneliness, and further analyzes the specific patterns of this relationship across different parent-adolescent gender dyads. This will deepen the understanding of parent–child interaction processes in specific gender contexts and test and enrich the transactional model of development. It will also provide more targeted practical implications for interventions targeting parental burnout and for the prevention of early adolescent loneliness.

1.1 Effects of Parental Burnout on Early Adolescent Loneliness

Parental burnout is a psychological syndrome that arises when parents endure chronic parenting stress over a prolonged period (Roskam et al., 2017, 2018), manifested as emotional exhaustion, fatigue in the parental role, emotional distancing from one's children, and a decline in parenting self-efficacy. Loneliness is a subjective experience in which individuals feel dissatisfied with their social relationships (Asher et al., 1984), and it is strongly influenced by social support (Salimi & Bozorgpour, 2012). Although the sources of adolescents' social support gradually shift from the family to peers, parental support still plays a crucial role during junior high school (Zou, 1999). Family factors such as parents therefore exert a non-negligible influence on the formation and development of children's loneliness. When parents experience burnout, they may evade their parental responsibilities and reduce emotional connection and support with their children, thereby increasing children's risk of loneliness. Moreover, burned-out parents may display hostility toward their children (Chen et al., 2022), and may even adopt negative parenting practices such as neglect and violence (Mikolajczak et al., 2018, 2019, 2023). Parenting practices in turn shape adolescents' ability to handle peer relationships: the way parents treat their children may be internalized by adolescents as a model for relating to peers (Bian et al., 2026). Specifically, after parents experience burnout they may adopt negative parenting practices, leaving adolescents unable to feel safe and confident in peer interactions and to encounter more difficulties in managing peer relationships, exhibiting social anxiety or social avoidance behaviors and further elevating early adolescent loneliness (Ayhan & Beyazit, 2021; Suárez-Relinque et al., 2023; Sun et al., 2024; Xu et al., 2017; Qin & Peng, 2024; Song & Hao, 2005). Adolescents may even resort to force when encountering peer conflict because of poor communication skills, further leading to future peer victimization and increased loneliness (Brinker et al., 2023; González-Abaurrea et al., 2026; Matthews et al., 2022; Ren et al., 2023; Zheng et al., 2025). Conversely, parents with lower levels of burnout possess a stronger sense of parenting efficacy and tend to adopt positive parenting practices such as warmth and autonomy support, which are associated with fewer early adolescent adjustment problems (Yang et al., 2021). This may be because children learn ways of conducting themselves from their parents' positive parenting behaviors, become more empathetic in handling peer relationships, get along better with peers, and maintain loneliness at a relatively low level (Bai et al., 2011; Wang et al., 2013). To date, however, only one cross-sectional study has supported the association between parental burnout and early adolescent loneliness (Cheng et al., 2020), and its longitudinal effects remain to be further tested. Accordingly, this study proposes

Hypothesis 1: parental burnout may longitudinally influence early adolescent loneliness.

1.2 Effects of Early Adolescent Loneliness on Parental Burnout

Owing to the strong societal concern for children's and adolescents' development and well-being, a large body of prior research has focused on the positive or negative influences of parents on children's development, neglecting the possibility that children may likewise influence their parents (Rueter & Conger, 1998). In fact, the emergence of behavioral, emotional, and other adjustment problems in children is also an important external trigger of parental stress and burnout (McSherry et al., 2019; Ren et al., 2024), which is related to parents needing to provide more attention, patience, and care (Blanchard et al., 2006). The additional parenting investment means that parents are more frequently placed in parenting-stress situations and gradually become trapped in parental burnout (Roskam et al., 2017). Adolescence is a period of "storm and stress." During this stage, adolescents' relational needs intensify and the risk of various adjustment problems such as loneliness increases (Qualter et al., 2015), and parents are more likely to face a combination of growing parenting demands and insufficient parenting resources. Neece et al. (2012) further pointed out that as individuals grow older, children's autonomy and agency gradually strengthen, making the role of their adjustment problems in predicting parenting stress more pronounced and stable. Empirical studies have examined the influence of children's and adolescents' adjustment problems on parental burnout (Hong et al., 2022; Minh et al., 2022), but the unique influence of early adolescent loneliness on parental burnout remains unclear. Given that adjustment problems often share comorbidity mechanisms (Grygiel et al., 2024; Yang et al., 2023; Geng & Tao, 2018), early adolescent loneliness may also predict parental burnout. On the one hand, children with higher levels of loneliness may refuse to communicate with their parents or even engage in frequent conflicts with them, making it difficult to establish close relationships with parents (Lodder et al., 2016); on the other hand, they may also exhibit externalizing behaviors, especially internet addiction and aggressive behavior (Check et al., 1985; Moretta & Buodo, 2020; Wang et al., 2025; Wang & Zeng, 2024; Peng et al., 2018). These factors may lead parents to worry continuously about adolescents' development, giving rise to parental emotional fatigue and exhaustion. More importantly, parents who receive no positive response over the long term may also withdraw emotionally and doubt their own parenting competence (Solholm et al., 2019), ultimately triggering parental burnout. Accordingly, this study proposes Hypothesis 2: early adolescent loneliness may also longitudinally influence parental burnout.

1.3 Reciprocal Relationships between Parental Burnout and Early Adolescent Loneliness and the Role of Parent-adolescent Gender Dyads

The transactional model of development (Sameroff, 1975; Sameroff & MacKenzie, 2003) offers a bidirectional framework for understanding parent–child interaction. The model emphasizes that individual development results from the continuous, dynamic interaction between the individual and the family and social environment. Parents and children constitute the most basic family unit, and each serves as a proximal environmental factor for the other's development. It is necessary to examine simultaneously how parental factors elicit specific responses in children (i.e., parent-driven effects) and how children's psychological and behavioral responses influence parents (i.e., child-driven effects). After reviewing nearly 15 years of research in the field of parental burnout, Mikolajczak et al. (2023) pointed out that the factors that trigger parental burnout may simultaneously become consequences of parental burnout, indicating reciprocal relationships. An empirical study of Western families by Woine et al. (2026) found a bidirectional association between parental burnout and children's and adolescents' adjustment problems. Whether such reciprocal relationships likewise exist in Chinese families, and whether they vary across parent-adolescent gender dyads, however, remain to be empirically tested.

The genders of parents and children may influence the strength of parent-driven and child-driven effects through the family division of labor, emotional expression styles, and parent–child interaction patterns, thereby shaping different influence patterns. First, regarding parental gender, according to the traditional division of labor in which "men manage the external affairs and women manage the internal affairs," fathers are more often assigned the responsibility of financial provision, whereas mothers undertake more of the daily care and emotional support duties (Jia & Ma, 2015; Wang, 2016). This division may cause mothers to deal more frequently with children's daily emotional and relational distress and make them more susceptible to the cumulative effects of parenting stress. Existing research has found that mothers report higher levels of parental burnout than fathers (de Santis et al., 2025). In addition, another study found a reciprocal relationship between mothers' internalizing problems and those of adolescents aged 14–18, but no reciprocal relationship between fathers' internalizing problems and those of adolescents aged 14–18 (Hughes & Gullone, 2010). This suggests that parental gender may affect parents' response sensitivity in bidirectional parent–child interactions. Second, regarding adolescent gender, boys and girls also differ in their response patterns in parent–child interactions. For example, among boys, a reciprocal relationship exists between poor parental monitoring and adolescents' negative emotions, whereas no such relationship is found among girls (Seely & Pössel, 2023). In a survey of Chinese elementary school students, Xing and Wang (2013) found that parental corporal punishment significantly and positively predicted the internalizing problems of both boys and girls, but only boys' internalizing problems were related to subsequent parental corporal punishment. This suggests that adolescent gender may likewise affect response sensitivity in bidirectional parent–child interactions.

Given that both parental gender and adolescent gender may influence bidirectional parent–child relationships, examining parental gender or adolescent gender only in isolation may be insufficient to reveal the distinct interaction patterns within specific parent–child dyads. In the field of parental burnout, existing studies of bidirectional parent–child relationships have attended to the roles of both parental and adolescent gender, providing indirect evidence. For example, in a study of early adolescents around age 13, Yang et al. (2025) found that parental burnout led to intensified mother–child conflict but did not significantly predict father–child conflict, whereas parent–child conflict jointly predicted subsequent increases in both fathers' and mothers' parental burnout; moreover, they found that the reciprocal relationship between parental burnout and parent–child conflict existed for both boys and girls. Thus, parental gender and adolescent gender may jointly shape the direction and strength of reciprocal relationships, but because that study did not directly compare the four parent-adolescent gender dyads—father–son, father–daughter, mother–son, and mother–daughter—it remains unable to determine whether different dyads exhibit unique reciprocal relationship patterns. Early research noted that, upon entering adolescence, girls increasingly engage in shared activities with their mothers and boys increasingly do so with their fathers, and this pattern is more pronounced in families with a younger opposite-gender sibling (Crouter et al., 1995). Livings (2021) further found that the effect of maternal depression on children's depression was stronger for daughters. In addition, research on parent–child interaction indicates that the effects of father–child and mother–child attachment on peer attachment exist for both boys and girls, but are stronger in same-gender parent–child dyads (i.e., father–son and mother–daughter) (Zou et al., 2020). These findings support the gender matching effect (Li & Meier, 2017), which is consistent with the explanations offered by gender socialization theory and social learning theory (Bandura, 1977; Fagot et al., 2000; Stockard, 2006). Particularly in adolescence, early adolescents begin to establish ego identity and are more susceptible to the pressure of gender norms concerning how boys or girls should behave; they then take the same-gender parent as a role model, learning that parent's gender-role experiences, emotional expression styles, and social adaptation behaviors, resulting in the phenomenon that parents exert stronger influences on the development of same-gender children than on opposite-gender children. Huston (1983) also emphasized that parents bear greater responsibility for the

socialization of same-gender children and may be more likely to promote the positive adjustment of same-gender children based on their own gender experiences. Conversely, when children's adjustment outcomes deviate from traditional gender-role expectations, same-gender parents may experience a lower sense of parenting efficacy. Therefore, when parent and child are gender-matched (i.e., father–son or mother–daughter), both parties are more likely to perceive and internalize each other's emotional states and behavior patterns, and their interactions may also be closer. To date, however, it remains unclear whether this gender matching effect also exists in the reciprocal relationship between parental burnout and early adolescent loneliness, and whether it is likewise reflected in child-driven effects. Examining this issue will help further clarify the differential roles of fathers and mothers in the emotional adjustment of boys and girls, reveal how early adolescents' emotional states in turn influence the parenting experiences of parents of different genders, and thereby clarify the specific manifestations of mutual parent–child influence across different parent-adolescent gender dyads, on the basis of which the boundary conditions of the transactional model of development can be tested and extended. It is therefore necessary to further consider the role of parent-adolescent gender dyads when examining the reciprocal relationships between parental burnout and early adolescent loneliness. In sum, this study proposes Hypothesis 3, that reciprocal relationships may exist between parental burnout and early adolescent loneliness, and Hypothesis 4, that the relationship patterns may differ across parent-adolescent gender dyads, with stronger effects in same-gender dyads.

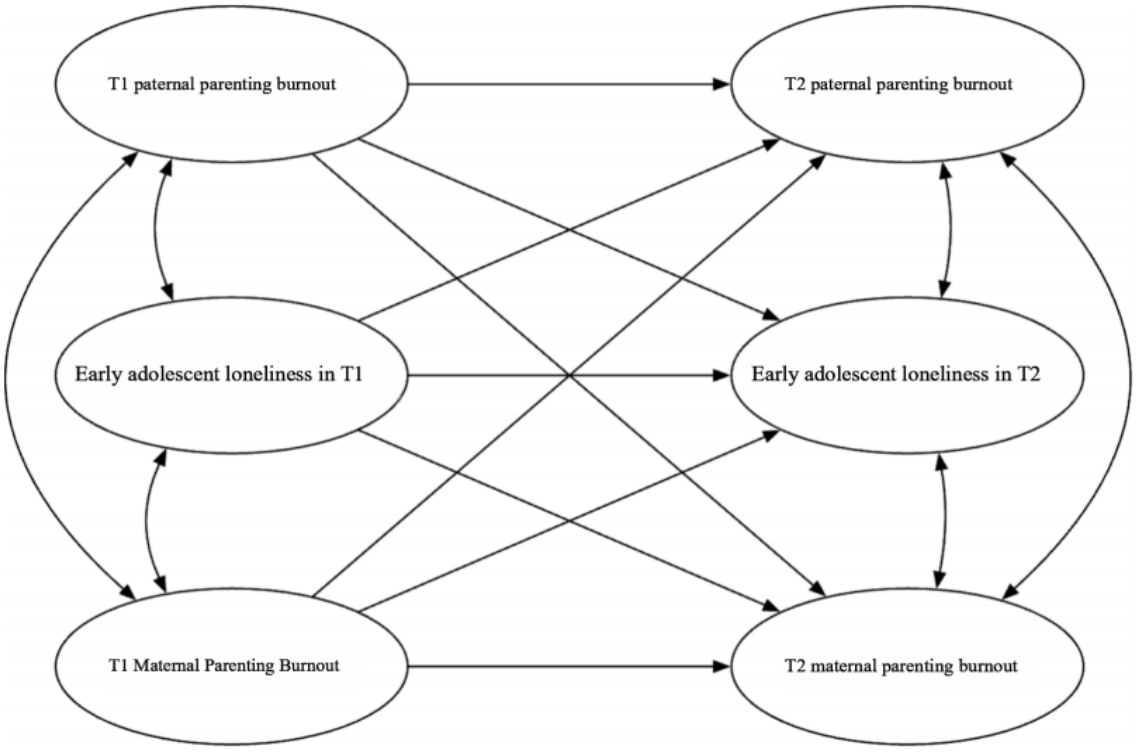


Figure 1 The hypothesized model to be tested

2 Methods

2.1 Participants

The participants were seventh-grade students and their parents from eight middle schools in Beijing and Hainan Province. Two waves of data collection were conducted: the first wave (T1) was carried out in June 2021, and after a one-year interval, the second wave (T2) was conducted in June 2022. Before each wave, informed consent was obtained from both the students and their parents. All participants were clearly informed of the purpose of the study and could withdraw unconditionally at any time without any

consequences. At T1, 1,571 complete sets of family data were collected. At T2, 518 fathers, 482 mothers, and 245 students failed to complete the corresponding questionnaires. Little's MCAR test supported a completely random attrition pattern ($\chi^2 = 39.541$, df = 30, $p = 0.114$). To further assess attrition bias, this study compared the attrition group and the retained group on the main demographic variables. The two groups did not differ significantly in adolescent gender, adolescent age, or father age ($p > 0.05$). A significant difference emerged in mother age ($t = -2.274$, $p = 0.023$): mothers in the attrition group (41.86 years) were significantly older than those in the retained group (41.31 years), with a very small effect size ($|d| = 0.125$). This effect size was below the reference standard for a small effect ($d = 0.20$) proposed by Cohen (1988), indicating that the practical significance of this difference was limited. Overall, the attrition did not show obvious systematic bias. The data of families that did not provide complete responses at T2 were therefore excluded. The final sample comprised 987 families, of which boys accounted for 46.61%. In Grade 7, the early adolescents included in the analysis had a mean age of 13.54 ± 0.34 years, fathers had a mean age of 43.34 ± 5.59 years, and mothers had a mean age of 41.31 ± 4.63 years. Table 1 presents the distribution of parental education in the included sample.

Table 1 Distribution of parental education

Education level	Fathers	Mothers
Junior high school or below	363 (36.8%)	405 (41.0%)
Senior high school / secondary vocational	190 (19.3%)	155 (15.7%)
Junior college	136 (13.8%)	138 (14.0%)
Bachelor's degree or above	291 (29.5%)	278 (28.2%)
Missing	7 (0.7%)	11 (1.1%)
Total	987 (100.0%)	987 (100.0%)

2.2 Measures

2.2.1 Parental burnout

This study used the Chinese version of the Parental Burnout Assessment (Cheng et al., 2020), the original English version of which was developed by Roskam et al. (2018). The scale contains 23 items and four dimensions. Exhaustion in the parenting role comprises 9 items, such as "I feel completely run down in my role as a parent." Contrast in parental self comprises 6 items, such as "I feel that I am no longer the good parent I used to be." Feelings of being fed up with parenting comprise 5 items, such as "I don't enjoy spending time with my children." Emotional distancing comprises 3 items, such as "I want to express my love to my children as I used to, but I feel unable to do so." All items were rated on a 7-point scale: 1 = never, 2 = a few times a year, 3 = once a month or less, 4 = a few times a month, 5 = once a week, 6 = a few times a week, and 7 = every day. Fathers and mothers separately reported their own levels of parental burnout, with higher scores indicating higher levels of burnout. The scale has shown good reliability and validity in previous studies (Bian et al., 2025). In this study, Cronbach's α for both fathers' and mothers' parental burnout was 0.98 at T1 and 0.99 at T2.

2.2.2 Early adolescent loneliness

This study used the Chinese version of the Children and Adolescents Loneliness Scale, developed by Asher et al. (1984) and later revised by the project team of the "Survey of Psychological Development Characteristics of Chinese Children and Adolescents Aged 6–15" (Dong & Lin, 2011). The revised scale has a unidimensional structure and contains 16 items, such as "I feel lonely," including 6 reverse-scored items such as "I can find friends when I need them." All items were rated on a 4-point scale: 0 = strongly disagree, 1 = somewhat disagree, 2 = somewhat agree, and 3 = strongly agree. After reverse scoring, higher scores

indicated higher levels of early adolescent loneliness. The scale has shown good reliability and validity in previous studies (Wang et al., 2026). In this study, Cronbach's α for early adolescent loneliness was 0.93 at both T1 and T2.

2.2.3 Control variable

Previous research has found that higher family socioeconomic status is associated with lower parental burnout and better adolescent mental health (Ren et al., 2024; Sorhagen & Wurster, 2017). It is therefore necessary to control for the potential influence of subjective family socioeconomic status. This study used the MacArthur scale, which assesses adolescents' subjectively perceived family socioeconomic status through ten rungs (Adler et al., 2000). The higher the rung, the higher the corresponding score, representing a higher perceived level of family education and social status.

2.3 Data analysis

SPSS was used for descriptive statistics and correlation analyses, and Mplus was used for the joint cross-group and longitudinal measurement invariance tests and for multigroup latent cross-lagged panel analyses. Joint cross-group and longitudinal measurement invariance refers to simultaneously testing, within a single multigroup longitudinal confirmatory factor analysis model, whether the measurement model remains consistent across the two dimensions of "group" and "time." Cross-group invariance (also termed multigroup invariance) concerns whether the same construct has the same factor structure and measurement meaning across different measurement objects, whereas longitudinal invariance (also termed temporal invariance) concerns whether the same construct has the same factor structure and measurement meaning across different time points. Satisfying joint cross-group and longitudinal measurement invariance ensures that, when subsequently comparing latent relationships across different parent–child dyads and time points, the same psychological construct is being measured in each group and at each time point. First, for parental burnout, the four dimensions of burnout were used as observed indicators to test the cross-group invariance of the measurement model across the four subgroups of father–son, father–daughter, mother–son, and mother–daughter, while also testing the longitudinal invariance of each subgroup across T1 and T2. Second, for early adolescent loneliness, the 16 items of the loneliness scale were used as observed indicators to test the cross-group invariance of the measurement model across children's genders, while also testing the longitudinal invariance of each gender group across T1 and T2. Nested model comparisons were used to evaluate changes in fit among the configural, metric, and scalar invariance models. If the decrease in CFI was less than 0.010 and the increase in RMSEA was less than 0.015 (Chen, 2007), the constrained model was accepted. When the measurement model reached scalar invariance, measurement invariance was considered established. Multigroup latent cross-lagged panel analyses were then conducted to examine the reciprocal relationships among fathers' burnout, mothers' burnout, and the loneliness of children of different genders, while controlling for subjective family socioeconomic status. Model fit was evaluated using χ^2/df, CFI, TLI, RMSEA, and SRMR. Following Wheaton et al. (1977), χ^2/df should be below 5. The other fit indices were judged against the cutoff criteria proposed by Hu and Bentler (1999): CFI and TLI no lower than 0.90, RMSEA no higher than 0.06, and SRMR no higher than 0.08. When these conditions were met, the model was considered to fit well.

3 Results

3.1 Descriptive statistics and correlation analysis

Table 2 presents the descriptive statistics and correlation analysis results for the main study variables. Both T1 fathers' and T1 mothers' parental burnout were significantly and positively correlated with T2 early adolescent loneliness. T1 early adolescent loneliness was significantly and positively correlated with T2

fathers' and T2 mothers' parental burnout. In addition, gender was significantly and negatively correlated with both T2 fathers' and T2 mothers' parental burnout: at T2, the fathers of girls reported lower burnout than the fathers of boys, and the mothers of girls reported lower burnout than the mothers of boys. Subjective family socioeconomic status was significantly and negatively correlated with fathers' and mothers' parental burnout and with early adolescent loneliness at both T1 and T2.

Table 2 Descriptive statistics and correlations among the main study variables

Variable		1	2	3	4	5	6	7	8
1 Gender		—							
2 Subjective family SES		—	—						
3 T1 Fathers' PB		0.02	−0.15***	—					
4 T2 Fathers' PB		−0.10**	−0.13***	0.40***	—				
5 T1 Mothers' PB		0.02	−0.13***	0.62***	0.29***	—			
6 T2 Mothers' PB		−0.10**	−0.17***	0.33***	0.62***	0.39***	—		
7 T1 Loneliness		0.05	−0.26***	0.18***	0.16***	0.17***	0.16***	—	
8 T2 Loneliness		0.02	−0.26***	0.13***	0.14***	0.17***	0.16***	0.53***	—
M ± SD (all)		—	5.79 ± 1.63	1.73 ± 1.11	2.01 ± 1.31	1.81 ± 1.18	1.98 ± 1.26	0.74 ± 0.59	0.73 ± 0.57
M ± SD (boys)		—	5.94 ± 1.65	1.71 ± 1.09	2.15 ± 1.43	1.78 ± 1.14	2.11 ± 1.40	0.71 ± 0.60	0.71 ± 0.58
M ± SD (girls)		—	5.67 ± 1.60	1.74 ± 1.13	1.89 ± 1.19	1.83 ± 1.21	1.87 ± 1.13	0.77 ± 0.59	0.74 ± 0.56

Note. *p < 0.05, **p < 0.01, ***p < 0.001. Gender was dummy-coded, with 1 = boys and 2 = girls. PB = parental burnout; SES = socioeconomic status.

3.2 Measurement invariance tests

Table 3 presents the results of the measurement invariance tests. The measurement model for parental burnout satisfied cross-group invariance across the four subgroups of father–son, father–daughter, mother–son, and mother–daughter, as well as the longitudinal invariance of each subgroup across T1 and T2. The measurement model for loneliness satisfied cross-group invariance across children's genders, as well as the longitudinal invariance of each gender group across T1 and T2.

Table 3 Results of measurement invariance tests

Variable	Model	χ^2	df	χ^2/df	CFI	RMSEA	$\Delta\chi^2$	ΔCFI	ΔRMSEA
Parental burnout	Model 1	480.740	180	2.671	0.987	0.058	—	—	—
	Model 2	603.728	208	2.903	0.983	0.062	122.988	−0.004	0.004
	Model 3	697.982	236	2.958	0.980	0.063	94.254	−0.003	0.001
Early adolescent loneliness	Model 1	2738.697	866	3.162	0.908	0.066	—	—	—
	Model 2	2876.747	914	3.147	0.903	0.066	138.050	−0.005	0.000
	Model 3	3060.054	962	3.181	0.897	0.067	183.307	−0.006	0.001

Note. Model 1 = configural (unconstrained) model; Model 2 = metric (weak invariance) model; Model 3 = scalar (strong invariance) model.

3.3 Multigroup latent cross-lagged panel analysis

To test whether the boys' and girls' groups differed in the patterns of the relationship between parental burnout and early adolescent loneliness, this study compared the unconstrained and constrained models. The unconstrained model allowed the parameters of the boys' and girls' groups to be estimated freely, whereas the constrained model imposed equality on the corresponding path coefficients and latent variable correlations of the two groups. The two models differed significantly (unconstrained model: $\chi^2 = 5145.687$, df = 2160; constrained model: $\chi^2 = 5181.180$, df = 2178; $\Delta\chi^2 = 35.493$, Δdf = 18, p = 0.008). This indicates

that constraining the corresponding path coefficients and latent correlations of the boys' and girls' groups to equality significantly worsened model fit, demonstrating a gender difference between the two groups in the patterns of the relationship between parental burnout and early adolescent loneliness. The boys' model and girls' model were then estimated separately to further present the specific relationship patterns in each group.

The fit results of the boys' model were as follows: $\chi^2(1080) = 2492.507$, $\chi^2/df = 2.308$; CFI = 0.928; TLI = 0.922; RMSEA = 0.054 (90% CI [0.052, 0.057]); SRMR = 0.064. Overall, the model fit well. The effect of subjective family socioeconomic status on T2 fathers' burnout was not significant ($\beta = -0.065$, $p = 0.151$), and its negative effects on T2 mothers' burnout were marginally significant ($\beta = -0.085$, $p = 0.062$; $\beta = -0.087$, $p = 0.056$). After controlling for subjective family socioeconomic status, fathers' burnout, mothers' burnout, and boys' loneliness all showed significant autoregressive effects between T1 and T2 ($\beta = 0.323$, $p < 0.001$; $\beta = 0.257$, $p < 0.001$; $\beta = 0.456$, $p < 0.001$). T1 boys' loneliness significantly and positively predicted T2 fathers' burnout ($\beta = 0.160$, $p = 0.001$) and T2 mothers' burnout ($\beta = 0.128$, $p = 0.007$). To compare whether the predictive effects of T1 boys' loneliness on T2 fathers' and mothers' burnout differed, a Wald test was conducted, which showed that the difference did not reach significance ($\chi^2(1) = 0.726$, $p = 0.394$). In addition, a positive effect of T1 fathers' burnout on T2 mothers' burnout was found ($\beta = 0.147$, $p = 0.008$). No other significant cross-lagged effects were found. Thus, Hypothesis 2 was supported, whereas Hypotheses 1, 3, and 4 were not supported.

The fit results of the girls' model were as follows: $\chi^2(1080) = 2653.180$, $\chi^2/df = 2.457$; CFI = 0.932; TLI = 0.926; RMSEA = 0.054 (90% CI [0.051, 0.056]); SRMR = 0.068. Overall, the model fit well. The effect of subjective family socioeconomic status on T2 fathers' burnout was not significant ($\beta = -0.052$, $p = 0.220$), while its negative effects on T2 mothers' burnout and T2 girls' loneliness were significant ($\beta = -0.127$, $p = 0.003$; $\beta = -0.137$, $p = 0.001$). After controlling for subjective family socioeconomic status, fathers' burnout, mothers' burnout, and girls' loneliness all showed significant autoregressive effects between T1 and T2 ($\beta = 0.409$, $p < 0.001$; $\beta = 0.357$, $p < 0.001$; $\beta = 0.420$, $p < 0.001$). T1 mothers' burnout significantly and positively predicted T2 girls' loneliness ($\beta = 0.185$, $p = 0.001$). No other significant cross-lagged effects were found. Thus, Hypothesis 1 was supported, Hypotheses 2 and 3 were not supported, and Hypothesis 4 was partially supported.

3.4 Robustness analysis

Whether adolescents live with their parents is an important family-structure factor in parent–child interaction, which may affect the direction and strength of the association between parental burnout and early adolescent loneliness by altering the frequency of daily parent–child contact, opportunities for emotional communication, and parents' awareness of their children's psychological states. To test whether the main results were influenced by parental cohabitation, after retaining only families in which adolescents lived with both parents (N = 932), the multigroup latent cross-lagged panel analyses were repeated (again controlling for subjective family socioeconomic status). Among boys, T1 boys' loneliness significantly and positively predicted T2 fathers' burnout ($\beta = 0.151$, $p = 0.002$) and T2 mothers' burnout ($\beta = 0.139$, $p = 0.004$), and T1 fathers' burnout significantly and positively predicted T2 mothers' burnout ($\beta = 0.153$, $p = 0.008$). Among girls, T1 mothers' burnout significantly and positively predicted T2 girls' loneliness ($\beta = 0.136$, $p = 0.022$), and T1 fathers' burnout significantly and positively predicted T2 mothers' burnout ($\beta = 0.120$, $p = 0.038$). These results indicate that the significance and direction of the parent–child cross-lagged paths in both the boys' and girls' models were consistent with those in the main analysis sample (N = 987), indicating that the core findings of this study are robust.

4 Discussion

Although Western studies have examined the reciprocal relationships between parental burnout and children's and adolescents' adjustment problems, the patterns of these relationships in the Chinese cultural context remain unclear. Meanwhile, existing research has paid relatively little attention to the role of different parent-adolescent gender dyads and has not fully revealed the complex interactions within the family. Addressing these gaps, this study thoroughly examined the reciprocal relationships between Chinese parents' parental burnout and early adolescent loneliness and further investigated the role of different parent-adolescent gender dyads. For both boys and girls, no reciprocal relationship was found between parental burnout and early adolescent loneliness. Instead, distinct unidirectional patterns were found for boys and girls: boys' loneliness influenced both fathers' and mothers' parental burnout, representing a son-driven effect, whereas a specific path in which mothers influenced daughters was found for girls, representing a mother-driven effect. These findings indicate that the relationship between parental burnout and early adolescent loneliness does not necessarily take the form of a universal, symmetric bidirectional pattern, which is inconsistent with findings from Western families (Woine et al., 2026). This does not mean that parental burnout and early adolescent loneliness are unrelated; rather, it indicates that their association may be highly specific to parent-adolescent gender dyads. Unlike Woine et al. (2026), who treated parents as a whole without distinguishing children's gender, this study distinguished fathers and mothers as well as boys and girls, thereby refining the potential reciprocal relationships into more specific gendered unidirectional patterns. In other words, parent-driven and child-driven effects do not necessarily co-occur within a single parent–child dyad but may be concentrated in different dyads. Thus, although the expected reciprocal relationships were not found, this study further highlights the important role of parent-adolescent gender dyads in identifying influence pathways within the family.

At T1, neither fathers' nor mothers' parental burnout differed significantly between families with boys and families with girls, whereas at T2, both the fathers and mothers of boys reported higher levels of burnout. This shift from no gender difference to the emergence of a gender difference tentatively suggests that parents raising boys may be more likely to experience increased burnout in the subsequent stage. Further cross-lagged analyses showed that boys' loneliness significantly predicted both fathers' and mothers' burnout; within the framework of child-driven effects, this finding supports both the gender matching effect and the cross-gender effect, although the difference between the two effects did not reach significance. In contrast, girls' loneliness did not significantly predict either fathers' or mothers' burnout; that is, no daughter-driven effect was observed. Similar gender differences have been reported in prior research: boys' emotional problems may elicit negative parenting behaviors, whereas the effects of girls' emotional problems are not significant (Seely & Pössel, 2023; Xing & Wang, 2013). Gender socialization theory emphasizes that boys should be outgoing, strong, and independent (Fagot et al., 2000; Stockard, 2006), whereas negative emotions such as depression and loneliness are regarded as inconsistent with traditional masculinity (Mokhwelepa & Sumbane, 2025; Nordin et al., 2024). When boys display loneliness and withdrawal, such behavior is more likely to be viewed as deviating from traditional masculinity and as less socially acceptable (Rubin & Coplan, 2004), and is more likely to arouse parents' concern about their adjustment. Although fathers bear more important responsibility for the socialization of sons (Huston, 1983), under the influence of traditional notions such as "son preference" and "raising sons for support in old age" (Zhou & Yuan, 2014), both fathers and mothers may pay greater attention to boys and invest more resources in them. Against this backdrop of high expectations and high investment, when boys face adjustment problems, the anxiety of both fathers and mothers may be triggered simultaneously, leading parents to display higher levels of burnout. In addition, the stigma attached to emotional vulnerability further weakens boys' willingness to seek help after experiencing negative emotions (Mokhwelepa & Sumbane, 2025; Nordin et al., 2024). In the absence of emotional expression and support-seeking, boys' loneliness may be more likely to manifest in the form of externalizing behaviors (Check et al., 1985; Moretta & Buodo, 2020; Wang et al., 2025; Wang & Zeng,

2024; Peng et al., 2018). For example, a longitudinal study of South Korean students aged 10 to 13 found that boys' internalizing problems elicited externalizing behaviors, whereas this phenomenon was not found among girls (Lee & Bukowski, 2012). Compared with internalizing problems such as loneliness, externalizing behaviors are more readily recognized by parents and are more likely to make both parents experience greater stress and emotional exhaustion in parenting. That is, boys' loneliness may also partly lead to increased parental burnout by eliciting externalizing behaviors.

Although no daughter-driven effect was found in the girls' sample, the results showed that mothers' parental burnout significantly and positively predicted daughters' loneliness, indicating a unidirectional influence path from mothers to daughters. This finding further emphasizes that, within specific parent-adolescent gender dyads, parent-driven and child-driven effects may take different forms; meanwhile, the result also supports the gender matching effect in parent–child interaction from the perspective of parent-driven effects, and similar patterns have been supported by existing research. For instance, the intergenerational transmission effect of maternal depression on children's depression is more pronounced for daughters (Livings, 2021). Inconsistent with the hypothesis, however, this study did not observe the same gender matching effect between fathers' burnout and boys' loneliness. On the contrary, Stone et al. (2016) found that higher parental stress was associated with subsequent internalizing problems among both boys and girls, without obvious gender differences. This may be because parental roles were not distinguished, masking the heterogeneity across different parent-adolescent gender dyads. By examining the roles of fathers and mothers separately, this study identified the specific pathway in which mothers influence daughters rather than fathers influence sons, thereby extending and refining the conclusions of existing research to some extent. Specifically, this differential result may be related to differences in both parental roles and children's gender. On the one hand, the traditional gender-role division of labor in which "men manage the external affairs and women manage the internal affairs" remains dominant in Chinese society, and mothers usually participate more in the child-rearing process (Jia & Ma, 2015; Wang, 2016). Therefore, when mothers experience burnout and reduce their emotional investment and interaction in child-rearing, children are more likely to perceive this change, which exerts a more pronounced influence on their psychological adjustment; compared with mothers, fathers may participate less in daily care, so the association between changes in fathers' parenting states and children's psychological adjustment may be relatively weak. Relevant empirical evidence supports this explanation: for example, Wang et al. (2021) found that fathers' parental burnout did not significantly predict early adolescent loneliness. Ping et al. (2022) found that mothers' parental burnout significantly and positively predicted early adolescent depression, whereas fathers' burnout did not significantly predict early adolescent depression. On the other hand, girls also exhibit higher rejection sensitivity than boys (Maiolatesi et al., 2022). Girls are therefore more likely to detect the reduction in emotional support after mothers experience burnout, which exerts a stronger influence on their social-emotional development. Conversely, boys may respond less strongly to changes in fathers' parenting states.

Using a two-wave cross-lagged design and based on rigorous tests of joint cross-group and longitudinal measurement invariance, this study systematically examined the reciprocal relationships between parental burnout and early adolescent loneliness in Chinese families and the role of parent-adolescent gender dyads. By including data from fathers, mothers, and early adolescents simultaneously, the study helps to provide a more comprehensive understanding of the mutual influence processes within the family system from a multi-agent perspective. On this basis, the study further revealed distinct patterns of influence across different parent-adolescent gender dyads, and the findings were robust. These findings add empirical evidence to cross-cultural research on parental burnout, extend prior research perspectives, and carry important theoretical and practical implications. Theoretically, this study incorporated parent-adolescent gender dyads and the gender matching effect into the analytical framework, further refining the boundary

conditions of parent-driven and child-driven effects in the transactional model of development. The study found that mutual parent–child influence has specific directionality and dyadic differences. Specifically, in the parent-driven direction, mothers' burnout significantly predicted only daughters' loneliness, consistent with the hypothesis that same-gender parent–child pairs share tighter psychological bonds. In the child-driven direction, boys' loneliness predicted both fathers' and mothers' burnout, but the difference between the predictive effects was not significant, indicating that in this study the child-driven effect did not show stronger same-gender influence. In sum, mutual parent–child influence is not determined solely by the gender of either the parent or the child, but depends on the specific parent-adolescent gender dyad. Nor is the gender matching effect universally present; its applicability depends on the joint action of the direction of influence and the parent-adolescent gender dyad. In practical terms, this study suggests that family education guidance should attend further to the differentiated needs brought about by parent-adolescent gender dyads. For families raising boys, boys' loneliness is one of the significant predictors of changes in parents' parenting states. Family education guidance should therefore help parents view the emotional changes of adolescent boys rationally and make them realize that loneliness may be a common stage-specific experience in the development of peer relationships and the exploration of ego identity in early adolescence. This will help parents respond to boys' emotional needs with greater acceptance and alleviate parental burnout caused by cognitive bias or the accumulation of parenting anxiety. For families raising girls, the study suggests that mothers' parenting states play a more prominent role in daughters' emotional adjustment. The strong emotional bond between mothers and daughters may increase the risk that the consequences of maternal burnout are transmitted to daughters, but may also provide an important basis for the protective effects of positive maternal parenting. Mothers often assume more responsibility in daily care and are therefore more likely to face the risks of parenting stress and emotional exhaustion. Family education guidance should therefore help mothers become more aware of their own parenting emotions, recognize the potential influence of their own states on daughters' psychological adjustment, and master effective emotion regulation and stress-coping strategies so as to reduce the risk that the consequences of burnout are transmitted to daughters. On this basis, mothers should also be guided to adopt warm, responsive positive parenting behaviors, thereby promoting daughters' emotional adjustment and fostering a virtuous cycle of interaction between mothers and daughters.

Despite the above strengths and implications, this study has several limitations that warrant further investigation in future research. First, the sample included only early adolescents after they entered Grade 7 and did not cover children and adolescents of other age groups; therefore, the age range of the current sample should be fully considered when generalizing the findings. Future research should include samples from a broader age range to further examine the stability and differences of the reciprocal relationships between parental burnout and adolescent loneliness across developmental stages. Second, although this study considered the influence of the key family-structure factor of whether adolescents lived with their parents, limited by data collection, it did not examine the role of the potential family-structure factor of whether children are only children. On the one hand, siblings can provide compensatory support to each other, alleviating individual loneliness (Milevsky, 2005). On the other hand, larger-scale PISA surveys have found that family size (number of siblings) is not associated with adolescent loneliness (Twenge et al., 2021), and the Chinese National Mental Health Development Report (2021–2022) (Fu et al., 2023) also shows that differences in loneliness across birth order are very weak ($\eta^2 = 0.001$). Given the inconsistency of existing research, future studies may include whether children are only children in the analysis to further examine the differences in the relationship between parental burnout and early adolescent loneliness between only-child and non-only-child families. Third, the relevant results still need to be further verified in longitudinal data with more waves. Although two-wave data help examine the temporal ordering among variables, they cannot fully reflect their dynamic processes. Future research may use data from three or more time points to portray

more systematically the dynamic development of the relationship between parental burnout and adolescent loneliness. Finally, this study mainly adopted a variable-centered perspective, which may mask the group heterogeneity of parental burnout. Future research may turn to person-centered approaches (such as latent profile analysis) to identify different subtypes of parental burnout and explore their differentiated effects on the social-emotional development of adolescents of different genders.

5 Conclusion

In sum, the main conclusions of this study are as follows: (1) boys' loneliness positively predicted both fathers' and mothers' parental burnout; (2) mothers' parental burnout positively predicted girls' loneliness; and (3) no reciprocal relationships were found between fathers' or mothers' parental burnout and boys' or girls' loneliness.

References

Adler, N. E., Epel, E. S., Castellazzo, G., & Ickovics, J. R. (2000). Relationship of subjective and objective social status with psychological and physiological functioning: Preliminary data in healthy, White women. Health Psychology, 19(6), 586–592.

Asher, S. R., Hymel, S., & Renshaw, P. D. (1984). Loneliness in children. Child Development, 55(4), 1456–1464.

Ayhan, A. B., & Beyazit, U. (2021). The associations between loneliness and self-esteem in children and neglectful behaviors of their parents. Child Indicators Research, 14(5), 1863–1879.

Bai, C., Zhang, D., Chen, R., & Gu, G. (2011). Effects of parenting styles on loneliness among middle school students. Chinese Journal of School Health, 32(4), 473–474.

Bandura, A. (1977). Social learning theory. Prentice-Hall.

Bian, H., Bian, Y., & Liang, L. (2025). The impact of parental burnout on family cohesion: The mediating roles of parent–child relationships and parental differences. Current Psychology, 44(10), 9435–9445.

Bian, H., Pu, Q., He, Y., Zhang, Y., & Chen, F. (2026). How does mothers' parental burnout longitudinally predict adolescent bullying perpetration and victimization? The mediating role of psychological control rather than autonomy support. Journal of Youth and Adolescence, 55(8), 2096–2112.

Bian, H., Zhuang, R., Wang, J., Bian, Y., & Liang, L. (2026). Parental burnout and adolescents' online gaming addiction: The roles of basic psychological need satisfaction, volitional control, and gender. Psychological Development and Education, 42(1), 110–120.

Blanchard, L. T., Gurka, M. J., & Blackman, J. A. (2006). Emotional, developmental, and behavioral health of American children and their families: A report from the 2003 National Survey of Children's Health. Pediatrics, 117(6), e1202–e1212.

Brinker, V., Dewald-Kaufmann, J., Padberg, F., & Reinhard, M. A. (2023). Aggressive intentions after social exclusion and their association with loneliness. European Archives of Psychiatry and Clinical Neuroscience, 273(5), 1023–1028.

Check, J. V. P., Perlman, D., & Malamuth, N. M. (1985). Loneliness and aggressive behaviour. Journal of Social and Personal Relationships, 2(3), 243–252.

Chen, B.-B., Qu, Y., Yang, B., & Chen, X. (2022). Chinese mothers' parental burnout and adolescents' internalizing and externalizing problems: The mediating role of maternal hostility. Developmental Psychology, 58(4), 768–777.

Chen, F. (2007). Sensitivity of goodness of fit indexes to lack of measurement invariance. Structural Equation Modeling: A Multidisciplinary Journal, 14(3), 464–504.

Cheng, H., Wang, W., Wang, S., Li, Y., Liu, X., & Li, Y. (2020). Validation of a Chinese version of the Parental Burnout Assessment. Frontiers in Psychology, 11, Article 321. https://doi.org/10.3389/fpsyg.2020.00321

Cohen, J. (1988). Statistical power analysis for the behavioral sciences (2nd ed.). Routledge.

Crouter, A. C., Manke, B. A., & McHale, S. M. (1995). The family context of gender intensification in early adolescence. Child Development, 66(2), 317–329.

de Santis, L., Noronha, A. P. P., Mikolajczak, M., & Roskam, I. (2025). A step forward in understanding gender differences in parental burnout: Parental involvement and emotion regulation. Journal of Child and Family Studies, 34(1), 95–108.

Dong, Q., & Lin, C. (Eds.). (2011). An introduction to the standardized tests of psychological development of Chinese children and adolescents. China: Science Press.

Fagot, B. I., Rodgers, C. S., & Leinbach, M. D. (2000). Theories of gender socialization. In T. Eckes & H. M. Trautner (Eds.), The developmental social psychology of gender (pp. 65–89). Lawrence Erlbaum Associates Publishers.

Fu, X., Zhang, K., Chen, X., & Chen, Z. (Eds.). (2023). Chinese national mental health development report (2021–2022). China: Social Sciences Academic Press.

Geng, M., & Tao, F. (2018). Comorbidity of children's internalizing and externalizing behaviors and its impact on future development. Chinese Journal of School Health, 39(4), 637–640.

González-Abaurrea, E., González-Cabrera, J., Ortega-Barón, J., Escortell, R., & Machimbarrena, J. M. (2026). Bidirectional relationships between peer victimization, loneliness and solitude in adolescence: A systematic review. Adolescent Research Review, 11(1), 195–247.

Grygiel, P., Dolata, R., Humenny, G., & Muszyński, M. (2024). Depressive symptoms and loneliness among early adolescents: A psychometric network analysis approach. Journal of Child Psychology and Psychiatry, 65(2), 199–214.

Hong, V. T. P., An, N. H., Thao, T. T. P., Thao, L. N., & Thanh, N. M. (2022). Behavior problems reduce academic outcomes among primary students: A moderated mediation of parental burnout and parents' self-compassion. New Directions for Child and Adolescent Development, (185–186), 27–42.

Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. Structural Equation Modeling, 6(1), 1–55.

Hughes, E. K., & Gullone, E. (2010). Reciprocal relationships between parent and adolescent internalizing symptoms. Journal of Family Psychology, 24(2), 115–124.

Huston, A. C. (1983). Sex-typing. In E. M. Hetherington (Ed.), Handbook of child psychology: Vol. 4. Socialization, personality, and social development (pp. 387–467). Wiley.

Jia, Y., & Ma, D. (2015). A multi-perspective examination of changes in gender concepts: Taking "men manage external affairs and women manage internal affairs" as an example. Collection of Women's Studies, (3), 29–36.

Laursen, B., & Collins, W. A. (2004). Parent–child communication during adolescence. In A. L. Vangelisti (Ed.), Handbook of family communication (pp. 333–348). Lawrence Erlbaum Associates Publishers.

Lee, E. J., & Bukowski, W. M. (2012). Co-development of internalizing and externalizing problem behaviors: Causal direction and common vulnerability. Journal of Adolescence, 35(3), 713–729.

Li, S., Zhang, S., & Yu, G. (2024). Effects of parental burnout on the mental health of primary and secondary school students: Evidence and educational countermeasures. Educational Science Research, (10), 12–19.

Li, X., & Meier, J. (2017). Father love and mother love: Contributions of parental acceptance to children's psychological adjustment. Journal of Family Theory & Review, 9(4), 459–490.

Livings, M. S. (2021). The gendered relationship between maternal depression and adolescent internalizing symptoms. Social Science & Medicine, 291, Article 114464. https://doi.org/10.1016/j.socscimed.2021.114464

Lodder, G. M. A., Goossens, L., Scholte, R. H. J., Engels, R. C. M. E., & Verhagen, M. (2016). Adolescent loneliness and social skills: Agreement and discrepancies between self-, meta-, and peer-evaluations. Journal of Youth and Adolescence, 45(12), 2406–2416.

Maiolatesi, A. J., Clark, K. A., & Pachankis, J. E. (2022). Rejection sensitivity across sex, sexual orientation, and age: Measurement invariance and latent mean differences. Psychological Assessment, 34(5), 431–442.

Matthews, T., Caspi, A., Danese, A., Fisher, H. L., Moffitt, T. E., & Arseneault, L. (2022). A longitudinal twin study of victimization and loneliness from childhood to young adulthood. Development and Psychopathology, 34(1), 367–377.

McSherry, D., Fargas Malet, M., & Weatherall, K. (2019). The Strengths and Difficulties Questionnaire (SDQ): A proxy measure of parenting stress. The British Journal of Social Work, 49(1), 96–115.

Mikolajczak, M., Aunola, K., Sorkkila, M., & Roskam, I. (2023). 15 years of parental burnout research: Systematic review and agenda. Current Directions in Psychological Science, 32(4), 276–283.

Mikolajczak, M., Brianda, M. E., Avalosse, H., & Roskam, I. (2018). Consequences of parental burnout: Its specific effect on child neglect and violence. Child Abuse & Neglect, 80, 134–145.

Mikolajczak, M., Gross, J. J., & Roskam, I. (2019). Parental burnout: What is it, and why does it matter? Clinical Psychological Science, 7(6), 1319–1329.

Milevsky, A. (2005). Compensatory patterns of sibling support in emerging adulthood: Variations in loneliness, self-esteem, depression and life satisfaction. Journal of Social and Personal Relationships, 22(6), 743–755.

Minh, T. N., Quang, A. M. T., & Ha, T. A. (2022). The influence of children's behavior problems on parents' psychological well-being: A serial mediation model of parental psychological control and parental burnout. Children and Youth Services Review, 134, Article 106366. https://doi.org/10.1016/j.childyouth.2022.106366

Mokhwelepa, L. W., & Sumbane, G. O. (2025). Men's mental health matters: The impact of traditional masculinity norms on men's willingness to seek mental health support: A systematic review of literature. American Journal of Men's Health, 19(3), Article 15579883251321670. https://doi.org/10.1177/15579883251321670

Moretta, T., & Buodo, G. (2020). Problematic internet use and loneliness: How complex is the relationship? A short literature review. Current Addiction Reports, 7(2), 125–136.

Neece, C. L., Green, S. A., & Baker, B. L. (2012). Parenting stress and child behavior problems: A transactional relationship across time. American Journal on Intellectual and Developmental Disabilities, 117(1), 48–66.

Nordin, T., Degerstedt, F., & Granholm Valmari, E. (2024). A scoping review of masculinity norms and their interplay with loneliness and social connectedness among men in Western societies. American Journal of Men's Health, 18(6), Article 15579883241304585. https://doi.org/10.1177/15579883241304585

Peng, Y., Zhu, L., & Wang, Z. (2018). Parental emotional expression and adolescents' problem behaviors: The multiple mediating effects of parent–child attachment and loneliness. Psychological Development and Education, 34(4), 504–512.

Ping, Y., Liu, T., Wang, W., & Li, Y. (2022). The mediating role of parental depression between parental burnout and children's depression. Chinese Journal of Clinical Psychology, 30(4), 964–967.

Qualter, P., Vanhalst, J., Harris, R., Van Roekel, E., Lodder, G., Bangee, M., … Verhagen, M. (2015). Loneliness across the life span. Perspectives on Psychological Science, 10(2), 250–264.

Qin, Y., & Peng, Y. (2024). Effects of parenting styles on junior high school students' social anxiety: The chain mediating roles of peer acceptance and rumination. Psychological Development and Education, 40(1), 103–113.

Ren, P., Wang, Y., Liang, Y., Li, S., & Wang, Q. (2023). Bidirectional relationship between bullying victimization and functions of aggression in adolescents: The mediating effect of teacher justice. Journal of Adolescence, 95(6), 1245–1257.

Ren, X., Cai, Y., Wang, J., & Chen, Q. (2024). A systematic review of parental burnout and related factors among parents. BMC Public Health, 24(1), Article 376. https://doi.org/10.1186/s12889-024-17829-y

Roskam, I., Brianda, M.-E., & Mikolajczak, M. (2018). A step forward in the conceptualization and measurement of parental burnout: The Parental Burnout Assessment (PBA). Frontiers in Psychology, 9, Article 758. https://doi.org/10.3389/fpsyg.2018.00758

Roskam, I., Raes, M.-E., & Mikolajczak, M. (2017). Exhausted parents: Development and preliminary validation of the Parental Burnout Inventory. Frontiers in Psychology, 8, Article 163. https://doi.org/10.3389/fpsyg.2017.00163

Rubin, K. H., & Coplan, R. J. (2004). Paying attention to and not neglecting social withdrawal and social isolation. Merrill-Palmer Quarterly, 50(4), 506–534.

Rueter, M. A., & Conger, R. D. (1998). Reciprocal influences between parenting and adolescent problem-solving behavior. Developmental Psychology, 34(6), 1470–1482.

Salimi, A., & Bozorgpour, F. (2012). Percieved social support and social-emotional loneliness. Procedia–Social and Behavioral Sciences, 69, 2009–2013.

Sameroff, A. J. (1975). Transactional models in early social relations. Human Development, 18(1–2), 65–79.

Sameroff, A. J., & MacKenzie, M. J. (2003). Research strategies for capturing transactional models of development: The limits of the possible. Development and Psychopathology, 15(3), 613–640.

Seely, H. D., & Pössel, P. (2023). Parenting behavior and adolescent affect: Bidirectionality and the role of gender. Journal of Family Issues, 44(5), 1321–1342.

Song, G., & Hao, B. (2005). A study of students' social avoidance behaviors, distress experiences, and their relationship with parenting styles. China Journal of Health Psychology, 13(2), 133–135.

Solholm, R., Bjørknes, R., & Jakobsen, R. (2019). Change in mothers' perception of well-being and distress following treatment of child conduct problems. Journal of Child and Family Studies, 28(4), 1059–1070.

Sorhagen, N. S., & Wurster, T. J. (2017). Income within context: Relative income matters for adolescent social satisfaction and mental health. Journal of Child Psychology and Psychiatry, 58(6), 736–743.

Stockard, J. (2006). Gender socialization. In J. S. Chafetz (Ed.), Handbook of the sociology of gender (pp. 215–227). Springer.

Stone, L. L., Mares, S. H. W., Otten, R., Engels, R. C. M. E., & Janssens, J. M. A. M. (2016). The co-development of parenting stress and childhood internalizing and externalizing problems. Journal of Psychopathology and Behavioral Assessment, 38(1), 76–86.

Suárez-Relinque, C., Del Moral, G., León-Moreno, C., & Callejas-Jerónimo, J. E. (2023). Emotional loneliness, suicidal ideation, and alexithymia in adolescents who commit child-to-parent violence. Journal of Interpersonal Violence, 38(3–4), 4007–4033.

Sun, Z., Ding, W., Li, J., Jiang, M., Li, W., & Xie, R. (2024). Why am I always lonely? The lasting impact of childhood harsh parental discipline on adolescents' loneliness. Family Relations, 73(3), 1933–1948.

Twenge, J. M., Haidt, J., Blake, A. B., McAllister, C., Lemon, H., & Le Roy, A. (2021). Worldwide increases in adolescent loneliness. Journal of Adolescence, 93, 257–269.

Wang, J., Yang, Y., Chen, Y., Lin, H., Wang, T., Wang, Z., … Fu, C. (2025). Loneliness, internalizing and externalizing problems, and suicidal ideation among Chinese adolescents: A longitudinal mediation analysis. Journal of Adolescent Health, 76(1), 96–104.

Wang, L., Zhang, H., & Bian, Y. (2026). The relationship between peer support at individual and group levels and adolescents' internalizing problems: A social network analysis approach. Journal of Psychological Science, 49(1), 105–120.

Wang, M., Fan, C., & Zhou, R. (2013). The relationship between parental emotional warmth and adolescent loneliness: A serial mediation analysis. Studies of Psychology and Behavior, 11(5), 654–659.

Wang, W., Wang, S., Cheng, H., Wang, Y., & Li, Y. (2021). The mediating role of fathers' parental burnout between parenting stress and adolescents' mental health. Chinese Journal of Clinical Psychology, 29(4), 858–861.

Wang, W. (2016). A study of the intra-household division of labor based on gender. Journal of Chongqing University (Social Science Edition), 22(5), 135–143.

Wang, Y., & Zeng, Y. (2024). Relationship between loneliness and internet addiction: A meta-analysis. BMC Public Health, 24(1), Article 858. https://doi.org/10.1186/s12889-024-18366-4

Wheaton, B., Muthén, B., Alwin, D. F., & Summers, G. F. (1977). Assessing reliability and stability in panel models. Sociological Methodology, 8, 84–136.

Woine, A., Dardier, V., Mikolajczak, M., & Roskam, I. (2026). Bidirectional and transactional effects between parental burnout and child internalizing and externalizing behaviors. Journal of Family Psychology, 40(3), 447–457.

Xing, X., & Wang, M. (2013). Sex differences in the reciprocal relationships between mild and severe corporal punishment and children's internalizing problem behavior in a Chinese sample. Journal of Applied Developmental Psychology, 34(1), 9–16.

Xu, J., Ni, S., Ran, M., & Zhang, C. (2017). The relationship between parenting styles and adolescents' social anxiety in migrant families: A study in Guangdong, China. Frontiers in Psychology, 8, Article 626. https://doi.org/10.3389/fpsyg.2017.00626

Yang, B., Chen, B.-B., Qu, Y., & Zhu, Y. (2021). Impacts of parental burnout on Chinese youth's mental health: The role of parents' autonomy support and emotion regulation. Journal of Youth and Adolescence, 50(8), 1679–1692.

Yang, B., Zhou, Z., Qu, Y., & Chen, B.-B. (2025). Is there a vicious cycle between parental burnout and parent–adolescent conflict? A three-wave within-family analytic approach. Family Process, 64(1), Article e70015. https://doi.org/10.1111/famp.70015

Yang, M., Wei, W., Ren, L., Pu, Z., Zhang, Y., Li, Y., … Wu, S. (2023). How loneliness linked to anxiety and depression: A network analysis based on Chinese university students. BMC Public Health, 23(1), Article 2499. https://doi.org/10.1186/s12889-023-17435-4

Zheng, M., Gao, Y., Li, J., & Liu, X. (2025). Longitudinal relationship between aggressive behavior and non-suicidal self-injury among adolescent boys and girls: The mediating role of peer victimization. Journal of Youth and Adolescence, 54(6), 1506–1519.

Zhou, Q., & Yuan, Y. (2014). A theoretical and empirical study of "son preference" in family basic education investment decisions. Population Journal, 36(3), 14–24.

Zou, H. (1999). The social support system and peer relationships of middle school students. Journal of Beijing Normal University (Social Science Edition), (1), 34–42.

Zou, S., Wu, X., & Li, X. (2020). Coparenting behavior, parent–adolescent attachment, and peer attachment: An examination of gender differences. Journal of Youth and Adolescence, 49(1), 178–191.

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Psychological Development and Education DOI:10.16187/j.cnki.issn1001-4918.2026.06.12

父母养育倦怠和早期青少年孤独感的双向关系:基于亲子性别组合的分析*

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摘 要:为探讨父母养育倦怠与早期青少年孤独感之间的双向关系以及亲子性别组合在其中的作用,本研究采用交叉滞后设计,分别在初一和初二阶段开展两波调查,采用养育倦怠量表分别测量父亲和母亲的养育倦怠,采用儿童青少年孤独感量表测量早期青少年孤独感。最终987个家族的数据纳入分析,初一时,早期青少年平均年龄为13.54±0.34岁,父亲平均年龄为43.34±5.59岁,母亲平均年龄为41.31±4.63岁。在控制主观家庭社会经济地位后,交叉滞后分析结果显示:在男生中,他们的孤独感能够正向预测父亲和母亲养育倦怠;在女生中,母亲养育倦怠能够正向预测其孤独感;此外,父亲和母亲养育倦怠与男女生孤独感之间均未发现双向关系。这些发现表明,父母养育倦怠与早期青少年孤独感之间的关系呈现出因亲子性别组合而异的单向预测模式。

关键词:父母养育倦怠;孤独感;双向关系;亲子性别组合;早期青少年

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1 引言

随着社会经济持续发展,中国父母日益重视对子女的科学养育与高质量陪伴。然而,更高的养育期待意味着父母需要投入更多的时间与情感资源。当养育需求与养育资源长期失衡时,父母容易产生养育倦怠。已有研究表明,父母养育倦怠不仅会损害父母自身心理健康,还可能引发青少年适应不良(Chen et al., 2022; Mikolajczak et al., 2018; Yang et al., 2021; 边昊天 等, 2026; 李森 等, 2024; 平一凡 等, 2022; 王玮 等, 2021)。青春期是情感联结需求的高峰期,也是孤独体验的敏感期(Qualter et al., 2015)。《中国国民心理健康发展报告(2021—2022)》显示,约有40%的青少年表示"经常感到孤独";尤其在青春期早期,个体进入初中后,孤独感水平明显上升,并在初一至初二阶段持续增加(傅小兰 等, 2023)。然而,养育倦怠的父母却可能减少对青少年的情感支持,加剧早期青少年的孤独体验(Cheng et al., 2020)。与此同时,养育青春期子女本身也是一项极具挑战性的任务。青春期个体情绪敏感、追求独立,亲子间互动减少、冲突增加(Laursen & Collins, 2004),这些特征也可能加重父母的挫败感与情绪耗竭。换言之,早期青少年的适应不良可能反过来引发父母养育倦怠(Hong et al., 2022; Minh et al., 2022; Ren et al., 2024)。

究竟是父母养育倦怠预测早期青少年孤独感,还是早期青少年孤独感预测父母养育倦怠,或者二者兼而有之?一项针对西方样本的研究指出,父母养育倦怠和儿童青少年的内外化问题之间可能存在双向关系(Woine et al., 2026)。然而,中国父母养育倦怠与早期青少年孤独感之间是否存在双向关系,尚缺乏直接的纵向实证证据。特别值得注意的是,性别不仅会影响家庭中的角色期待,也会影响情感表达与沟通方式,因而不同的亲子组合之间可能存在差异化的互动过程。这意味着父母养育倦怠和早期青少年孤独感之间的关系模式也可能因亲子性别组合不同而有所差异,但已有研究对这一重要过程的理解相对有限。综上所述,本研究拟基于两波纵向数据,采用交叉滞后设计,探讨中国父母养育倦怠与早期青少年孤独感之间的双向关系,并进一步分析该关系在不同亲子性别组合中的具体模式。这有助于深化对特定性别情境下亲子互动过程的认

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识,检验并丰富发展相互作用模型。同时,本研究也能为父母养育倦怠的干预及早期青少年孤独感的预防提供更具针对性的实践启示。

1.1 父母养育倦怠对早期青少年孤独感的影响

养育倦怠是父母因长期承受慢性育儿压力而产生的一种心理综合征(Roskam et al., 2017; Roskam et al., 2018),表现为情感耗竭、对父母角色感到疲惫、与子女情感疏远以及养育效能感下降。孤独感作为一种个体对自身社交关系感到不满的主观体验(Asher et al., 1984),受社会支持的影响很大(Salimi & Bozorgpour, 2012)。虽然青少年社会支持的来源逐渐由家庭转向同伴,但在初中阶段,父母支持仍起到至关重要的作用(邹泓, 1999)。因此,父母等家庭因素对子女孤独感的形成与发展具有不可忽视的重要影响。当父母陷入养育倦怠时,他们可能逃避履行父母职责,减少与子女的情感联结和情感支持,从而增加子女产生孤独感的风险。此外,养育倦怠的父母可能对孩子表现出敌意(Chen et al., 2022),甚至采取忽视和暴力等消极教养方式(Mikolajczak et al., 2018; Mikolajczak et al., 2019; Mikolajczak et al., 2023)。而父母教养方式进一步塑造了青少年处理同伴关系的能力,即父母对待孩子的方式可能会被青少年内化为与同伴相处的模式(Bian et al., 2026)。具体而言,父母陷入养育倦怠后可能采取消极教养方式,导致青少年无法在同伴互动中感到安全和自信,在处理同伴关系时遇到更多困难,表现出社交焦虑或社交回避行为,进一步导致早期青少年孤独感水平上升(Ayhan & Beyazit, 2021; Suárez - Relinque et al., 2023; Sun et al., 2024; Xu et al., 2017; 秦瑶, 彭运石, 2024; 宋广文, 郝丙辉, 2005)。青少年甚至可能在遇到同伴矛盾时因不善沟通而诉诸武力,进一步导致未来的同伴受害和孤独感增加(Brinker et al., 2023; González - Abaurrea et al., 2026; Matthews et al., 2022; Ren et al., 2023; Zheng et al., 2025)。相反,养育倦怠水平更低的父母拥有更高的养育效能感,倾向于采用积极的教养方式,例如温暖、自主支持等,这与更少的早期青少年适应问题有关(Yang et al., 2021)。这可能是因为子女在父母的积极教养行为中习得处世之道,在处理同伴关系时可能更善解人意,与同伴相处更加融洽,早期青少年孤独感能够保持在较低水平(白春玉 等, 2011; 王明忠等, 2013)。然而,到目前为止,仅有一项横断研究支持了父母养育倦怠和早期青少年孤独感之间的关联(Cheng et al., 2020),纵向影响有待进一步检验。综上,本研究提出假设 1,父母养育倦怠可能纵向影响早期青少年孤独感。

1.2 早期青少年孤独感对父母养育倦怠的影响

由于社会对儿童青少年发展和福祉的高度关注,以往大量研究集中探讨父母对儿童青少年发展的积极或消极影响,忽略了子女可能同样对父母产生影响(Rueter & Conger, 1998)。事实上,孩子出现行为、情绪等适应问题也是引发父母压力和养育倦怠的重要外部动因(McSherry et al., 2019; Ren et al., 2024),这与父母需要给予更多关注、付出更多耐心并提供更多照顾有关(Blanchard et al., 2006)。额外的养育投入意味着父母更频繁地处于养育压力情境中,并逐渐陷入养育倦怠的困境(Roskam et al., 2017)。青春期是青少年发展的“疾风骤雨期”。在此阶段,青少年的关系需求增强,出现孤独感等各类适应问题的风险加剧(Qualter et al., 2015),父母更可能面临养育需求增加和养育资源不足的情况。Neece 等人(2012)进一步指出,随着个体年龄的增长,孩子的自主性和行动力会逐渐增强,这使得他们的适应问题在预测父母养育压力方面的作用更为显著和稳定。目前已有实证研究考察了儿童和青少年适应问题对父母养育倦怠的影响(Hong et al., 2022; Minh et al., 2022),但早期青少年孤独感对父母养育倦怠的独特影响仍不清楚。鉴于适应问题之间往往存在共病机制(Grygiel et al., 2024; Yang et al., 2023; 耿梦龙, 陶芳标, 2018),早期青少年的孤独感也可能预测父母养育倦怠。一方面,孤独水平较高的孩子可能拒绝与父母沟通,甚至与父母发生频繁冲突,难以与父母建立亲密关系(Lodder et al., 2016);另一方面,他们还可能表现出外化行为,尤其是网络成瘾和攻击行为(Check et al., 1985; Moretta & Buodo, 2020; Wang et al., 2025; Wang & Zeng, 2024; 彭源 等, 2018)。这些原因可能导致父母持续担忧青少年的发展状况,引发父母的情感疲惫和耗竭。更重要的是,长期得不到积极回应的父母也可能在情感上退缩,并对自身养育能力产生怀疑(Solholm et al., 2019),最终引发父母养育倦怠。综上,本研究提出假设 2,早期青少年孤独感也可能纵向影响父母养育倦怠。

1.3 父母养育倦怠和早期青少年孤独感的双向关系以及亲子性别组合的作用

发展相互作用模型(Sameroff, 1975; Sameroff & MacKenzie, 2003)提出了理解亲子互动的双向框

架。该模型强调,个体发展是个体与家庭及社会环境持续、动态互动的结果。父母和子女构成了最基本的家庭单位,他们互为对方发展的近端环境因素。有必要同时考察父母因素如何引发子女的特定反应(即父母驱动效应)以及子女的心理与行为反应如何影响父母(即子女驱动效应)。Mikolajczak 等人(2023)在回顾近 15 年来养育倦怠领域的研究后指出,引发父母养育倦怠的因素可能同时成为父母养育倦怠的后果,即存在双向关系。Woine 等人(2026)针对西方家庭的实证研究发现,父母养育倦怠和儿童青少年的适应问题之间存在双向关联。然而,这种双向关系在中国家庭中是否同样存在,以及是否会因亲子性别组合而异,仍有待实证检验。

父母和子女的性别可能通过家庭角色分工、情感表达方式以及亲子互动模式,影响父母驱动效应和子女驱动效应的强弱,进而塑造不同的影响模式。首先,从父母性别来看,根据“男主外、女主内”的传统分工模式,父亲更多被赋予经济供给责任,母亲则更多承担日常照料与情感支持职责(贾云竹, 马冬玲, 2015; 王玮玲, 2016)。这种分工可能使母亲更频繁地应对子女日常的情绪与关系困扰,也使其更容易受到养育压力的累积影响。已有研究发现,母亲的养育倦怠水平高于父亲(de Santis et al., 2025)。此外,另有研究发现,母亲的内化问题与 14 ~ 18 岁青少年的内化问题之间存在双向关系,但未发现父亲的内化问题与 14 ~ 18 岁青少年的内化问题之间存在双向关系(Hughes & Gullone, 2010)。这提示父母性别可能影响其在亲子双向互动中的反应敏感性。其次,从青少年性别来看,男生和女生在亲子互动中的反应模式上也存在差异。例如,在男生中,父母监管不力与青少年负面情绪之间存在双向关系,但在女生中则未发现这种关系(Seely & Pössel, 2023)。Xing 和 Wang(2013)对中国小学生的调查发现,父母体罚能显著正向预测男女生的内化问题,但只发现男生的内化问题与随后的父母体罚有关。这提示青少年性别同样可能影响其在亲子双向互动中的反应敏感性。

鉴于父母性别和青少年性别均可能影响亲子双向关系,仅分别考察父母性别或青少年性别,可能不足以揭示特定亲子组合中的差异化互动模式。在养育倦怠领域,现有亲子双向关系的研究已同时关注父母性别与青少年性别的作用,提供了间接证据。例如,Yang 等人(2025)针对 13 岁左右早期青少年开展的研究发现,父母养育倦怠会导致母子冲突加剧,而对父子冲突没有显著预测作用,亲子冲突则共同预示着未来父亲和母亲养育倦怠水平的上升;此外,他们还发现,父母养育倦怠和亲子冲突之间的双向关系在男女生中均存在。由此,父母性别与青少年性别可能共同塑造双向关系的方向与强度,但由于该研究并未直接比较父亲 - 儿子、父亲 - 女儿、母亲 - 儿子和母亲 - 女儿四类亲子性别组合的作用,因此仍无法说明不同亲子性别组合之间是否具有独特的双向关系模式。早期研究指出,进入青春期阶段,女孩越来越多地与母亲共同活动,男孩越来越多地与父亲共同活动,并且这种模式在有异性年幼同胞的家庭里更明显(Crouter et al., 1995)。Livings(2021)进一步发现,母亲抑郁对子女抑郁的影响在女儿中更强。此外,亲子互动方面的研究则指出,父子依恋和母子依恋对同伴依恋的影响在男生和女生群体中均存在,但在同性亲子组合(即父子、母女)中的效应更强(Zou et al., 2020)。这些发现支持了性别匹配效应(Li & Meier, 2017),这一效应与性别社会化理论以及社会学习理论的解释相符(Bandura, 1977; Fagot et al., 2000; Stockard, 2006)。特别是在青春期,早期青少年开始建立自我同一性,更容易受到“男生应当如何表现”或“女生应当如何表现”等性别规范压力的影响,进而将同性别的父母作为模仿对象,学习其性别角色经验、情绪表达方式和社会适应行为,表现出父母对同性子女发展的影响强于异性子女的现象。Huston(1983)也强调,父母对同性子女的社会化负有更大的责任,父母也可能更容易依据自身性别经验促进同性别子女的积极适应。反过来,当子女的适应结果偏离传统性别角色期待时,同性别的父母可能产生更低的养育效能感。因此,当亲子性别匹配(即父子或母女)时,亲子双方更容易感知并内化彼此的情绪状态和行为模式,双方之间的互动也可能更为密切。但到目前为止,尚不清楚这种性别匹配效应是否也存在于父母养育倦怠与早期青少年孤独感的双向关系中,以及是否同样体现在子女驱动效应中。对这一问题的考察有助于进一步明确父亲与母亲在男生和女生情绪适应中的差异化作用,揭示早期青少年的情绪状态如何反过来影响不同性别父母的养育体验,从而厘清不同亲子性别组合中亲子相互影响的具体表现,并在此基础上检验和拓展发展相互作用模型的适用边界。因此,在探讨父母养育倦怠与早期青少年孤独感的双向关系时,有必要进一步考虑亲子性别组合的作用。综上,本研究提出假设 3,父母养育倦怠和早期青少年孤独感之间可能

存在双向关系;假设 4,关系模式可能因亲子性别组合而不同,且在同性亲子组合中效应更强。

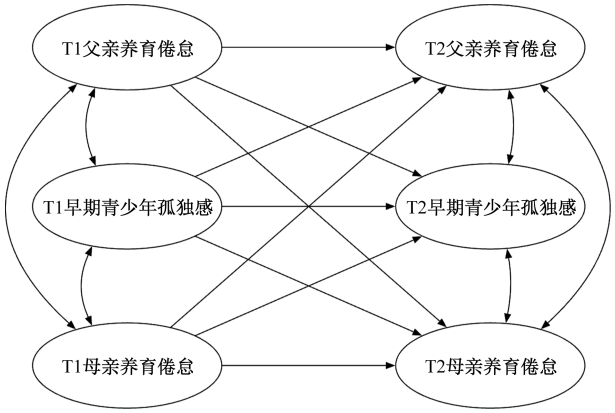


图 1 拟检验的假设模型

2 研究方法

2.1 被试

本研究以北京市和海南省 8 所中学的初一年级学生及其父母为研究对象,进行了两波数据收集,第一波调查(T1)于 2021 年 6 月开展;间隔一年后,于 2022 年 6 月进行了第二波调查(T2)。在每波调查开始前,均获得了学生及其父母的知情同意。所有参与者均被明确告知研究的目的,他们可以在任何时候无条件退出研究,且不会因此承担任何后果。T1 阶段共回收 1571 组完整的家庭数据。在 T2 阶段,共有 518 名父亲、482 名母亲及 245 名学生未能完成相应问卷。Little's MCAR 检验结果支持完全随机流失模式($\chi^2 = 39.541$, $df = 30$, $p = 0.114$)。为进一步评估样本流失偏差,本研究比较了流失组与保留组在主要人口学变量上的差异。结果显示,两组在青少年性别、青少年年龄和父亲年龄方面均不存在显著差异($p > 0.05$);在母亲年龄方面存在显著差异($t = -2.274$, $p = 0.023$),流失组母亲的年龄(41.86 岁)显著大于保留组(41.31 岁),效应量很小($|d| = 0.125$)。该效应量低于 Cohen(1988)提出的小效应参考标准($d = 0.20$),提示该差异的实际意义有限。综上,样本流失未表现出明显的系统性偏差。因此,本研究剔除了 T2 阶段未完整作答的家庭数据。最终共纳入 987 个家庭的数据,其中男生占比 46.61%。在初一时,纳入分析的早期青少年平均年龄为 13.54 ± 0.34 岁,父亲平均年龄为 43.34 ± 5.59 岁,母亲平均年龄为 41.31 ± 4.63 岁。表 1 呈现了纳入样本的父母受教育程度分布情况。

表 1 父母受教育程度分布情况

受教育程度	父亲	母亲
初中及以下	363(36.8%)	405(41.0%)
高中/中职	190(19.3%)	155(15.7%)
大专	136(13.8%)	138(14.0%)
本科及以上	291(29.5%)	278(28.2%)
缺失	7(0.7%)	11(1.1%)
合计	987(100.0%)	987(100.0%)

2.2 测量

2.2.1 父母养育倦怠

本研究采用中文版养育倦怠量表(Cheng et al., 2020),其原始英文版由 Roskam 等人(2018)开发。该量表共 23 个题目,包含四个维度:养育疲惫,包含 9 个题目,如"为了扮演好父母的角色,我已经累得筋疲力尽了";自我对比,包含 6 个题目,如"我觉得自己不再像以前一样是个好父母了";养育厌倦,包含 5 个题目,如"我并不享受和我的孩子共度的时光";情感疏远,包含 3 个题目,如"我想像过去一样向孩子们表达我对他们的爱,但感觉有心无力"。所有题目均采用 7 点计分,1 = 从不,2 = 一年几次,3 = 一个月一次或更少,4 = 一个月几次,5 = 一周一次,6 = 一周几次,7 = 每天。父亲和母亲分别报告自身的养育倦怠水平,得分越高表示养育倦怠水平越高。该量表在以往研究中表现出良好的信效度(Bian et al., 2025)。本研究中 T1 父亲和母亲养育倦怠的 Cronbach's α 系数均为 0.98,T2 父亲和母亲养育倦怠的 Cronbach's α 系数均为 0.99。

2.2.2 早期青少年孤独感

本研究采用由 Asher 等人(1984)开发,后经"中国 6~15 岁儿童青少年心理发育特征调查"项目组修订的中文版儿童青少年孤独感量表(董奇, 林崇德, 2011)。修订后的量表具有单维结构,包含 16 个题目,如"我感到寂寞",其中包含 6 个反向题目,如"当我需要时我可以找到朋友"。所有题目均采用 4 点计分,0 = 很不符合,1 = 不太符合,2 = 基本符合,3 = 很符合。反向题目经反向计分后,得分越高表示早期青少年的孤独感水平越高。该量表在以往研究中表现出良好的信效度(王凌飞 等, 2026)。本研究中 T1 和 T2 早期青少年孤独感的 Cronbach's α 系数均为 0.93。

2.2.3 控制变量

以往研究发现,更高的家庭社会经济地位与更

低的父母养育倦怠和更高的青少年心理健康水平有关(Ren et al., 2024; Sorhagen & Wurster, 2017)。因此,有必要控制主观家庭社会经济地位的潜在影响。本研究采用 MacArthur 量表,通过十个层级来评估青少年主观感知的家庭社会经济地位(Adler et al., 2000)。阶梯层级越高,对应的分数也越高,代表其感知到的家庭受教育水平及社会地位越高。

2.3　数据分析

本研究使用 SPSS 进行描述性统计和相关分析,使用 Mplus 进行跨主体与跨时间的联合测量不变性检验和多组潜变量交叉滞后分析。跨主体与跨时间的联合测量不变性是指在同一个多组纵向验证性因子分析模型中,同时检验测量模型是否在"主体"和"时间"两个维度上保持一致。跨主体不变性(也称多组不变性)关注同一构念在不同测量对象中是否具有相同的因子结构和测量含义;跨时间不变性(也称纵向不变性)关注同一构念在不同时间点上是否具有相同的因子结构和测量含义。满足跨主体与跨时间的联合测量不变性能够确保后续比较不同亲子组合以及不同时间点上的潜变量关系时,各组和各时间点所测量的是同一心理构念。首先,针对父母养育倦怠,以养育倦怠的四个维度作为观测指标,检验测量模型在父亲 - 儿子、父亲 - 女儿、母亲 - 儿子、母亲 - 女儿四个亚组之间的跨主体不变性,同时检验各亚组在 T1 和 T2 两个时间点上的纵向不变性。其次,针对早期青少年孤独感,以孤独感量表的 16 个题项作为观测指标,检验测量模型在子女不同性别之间的跨主体不变性,同时检验子女不同性别组在 T1 和 T2 两个时间点上的纵向不变性。采用嵌套模型比较方法,评估无约束模型、弱等值模型、强等值模型之间拟合度的变化。若 CFI 的降低量小于 0.010 且 RMSEA 的增加量小于 0.015(Chen, 2007),则接受约束模型。当测量模型达到强等值时,即认为测量不变性成立。随后采用多组潜变量交叉滞后分析,考察父亲养育倦怠、母亲养育倦怠以及不同性别子女孤独感之间的双向关系,同时对主观家庭社会经济地位予以控制。使用以下指标评估模型拟合度:χ^2/df、CFI、TLI、RMSEA 和 SRMR。根据 Wheaton 等人(1977)的建议,χ^2/df 应小于 5。其他拟合指数参照 Hu 和 Bentler(1999)提出的临界标准:CFI 与 TLI 不低于 0.90,RMSEA 不超过 0.06,SRMR 不超过 0.08。若满足上述条件,则认为模型拟合良好。

3　结果

3.1　描述性统计与相关分析

表 2 呈现了主要研究变量的描述性统计与相关分析结果。结果表明,T1 父亲养育倦怠和 T1 母亲养育倦怠均与 T2 早期青少年孤独感存在显著正相关。T1 早期青少年孤独感则与 T2 父亲养育倦怠和 T2 母亲养育倦怠存在显著正相关。此外,性别与 T2 父亲养育倦怠和 T2 母亲养育倦怠存在显著负相关,即 T2 时,女生父亲的养育倦怠水平低于男生父亲,女生母亲的养育倦怠水平低于男生母亲。主观家庭社会经济地位与 T1 和 T2 的父亲养育倦怠、母亲养育倦怠以及早期青少年孤独感都存在显著负相关。

3.2　测量不变性检验

表 3 呈现了测量不变性检验的结果。养育倦怠的测量模型同时满足父亲 - 儿子、父亲 - 女儿、母亲 -

表 2　主要研究变量的描述性统计与相关分析

变量	1	2	3	4	5	6	7	8
1 性别	—							
2 主观家庭社会经济地位	—	—						
3 T1 父亲养育倦怠	0.02	-0.15***	—					
4 T2 父亲养育倦怠	-0.10**	-0.13***	0.40***	—				
5 T1 母亲养育倦怠	0.02	-0.13***	0.62***	0.29***	—			
6 T2 母亲养育倦怠	-0.10**	-0.17***	0.33***	0.62***	0.39***	—		
7 T1 早期青少年孤独感	0.05	-0.26***	0.18***	0.16***	0.17***	0.16***	—	
8 T2 早期青少年孤独感	0.02	-0.26***	0.13***	0.14***	0.17***	0.16***	0.53***	—
$M \pm SD$(全体)	—	5.79 ± 1.63	1.73 ± 1.11	2.01 ± 1.31	1.81 ± 1.18	1.98 ± 1.26	0.74 ± 0.59	0.73 ± 0.57
$M \pm SD$(男生)	—	5.94 ± 1.65	1.71 ± 1.09	2.15 ± 1.43	1.78 ± 1.14	2.11 ± 1.40	0.71 ± 0.60	0.71 ± 0.58
$M \pm SD$(女生)	—	5.67 ± 1.60	1.74 ± 1.13	1.89 ± 1.19	1.83 ± 1.21	1.87 ± 1.13	0.77 ± 0.59	0.74 ± 0.56

注:*$p<0.05$, **$p<0.01$, ***$p<0.001$;性别虚拟编码,1 代表男生,2 代表女生。

表 3 测量不变性检验结果

变量	模型	χ^2	df	χ^2/df	CFI	RMSEA	$\Delta\chi^2$	ΔCFI	ΔRMSEA
父母养育倦怠	Model 1	480.740	180	2.671	0.987	0.058	—	—	—
	Model 2	603.728	208	2.903	0.983	0.062	122.988	−0.004	0.004
	Model 3	697.982	236	2.958	0.980	0.063	94.254	−0.003	0.001
早期青少年孤独感	Model 1	2738.697	866	3.162	0.908	0.066	—	—	—
	Model 2	2876.747	914	3.147	0.903	0.066	138.050	−0.005	0.000
	Model 3	3060.054	962	3.181	0.897	0.067	183.307	−0.006	0.001

注:Model 1 代表无约束模型;Model 2 代表弱等值模型;Model 3 代表强等值模型。

儿子、母亲 – 女儿四个亚组之间的跨主体不变性,以及各亚组在 T1 和 T2 两个时间点上的纵向不变性。孤独感的测量模型同时满足不同子女性别之间的跨主体不变性,以及不同子女性别组在 T1 和 T2 两个时间点上的纵向不变性。

3.3 多组潜变量交叉滞后分析

为检验男生组和女生组在父母养育倦怠与早期青少年孤独感之间的关系模式上是否存在差异,本研究比较了无约束模型与约束模型。无约束模型允许男生组和女生组的参数分别自由估计;约束模型则将两组对应的路径系数与潜变量相关系数约束为相等。结果显示,无约束模型与约束模型存在显著差异($\chi^2_{\text{无约束模型}} = 5145.687$, $df = 2160$; $\chi^2_{\text{约束模型}} = 5181.180$, $df = 2178$; $\Delta\chi^2 = 35.493$, $\Delta df = 18$, $p = 0.008$)。该结果表明,将男生组和女生组对应的路径系数与潜变量相关系数约束为相等后,模型拟合显著变差,说明两组在父母养育倦怠与早期青少年孤独感之间的关系模式上存在性别差异。本研究随后分别估计男生模型和女生模型,以进一步呈现两组中父母养育倦怠与早期青少年孤独感之间的具体关系模式。

男生模型的拟合结果如下:χ^2(1080) = 2492.507, $\chi^2/df = 2.308$; CFI = 0.928; TLI = 0.922; RMSEA = 0.054 (90% CI [0.052, 0.057]); SRMR = 0.064。整体来看,模型拟合良好。主观家庭社会经济地位对 T2 父亲养育倦怠的预测作用不显著($\beta = -0.065$, $p = 0.151$),对 T2 男生孤独感和 T2 母亲养育倦怠的负向预测作用边缘显著($\beta = -0.085$, $p = 0.062$; $\beta = -0.087$, $p = 0.056$)。在控制了主观家庭社会经济地位后,父亲养育倦怠、母亲养育倦怠及男生孤独感在 T1 与 T2 之间均呈现出显著的自回归效应($\beta = 0.323$, $p < 0.001$; $\beta = 0.257$, $p < 0.001$; $\beta = 0.456$, $p < 0.001$)。T1 男生孤独感能够显著正向预测 T2 父亲养育倦怠($\beta = 0.160$, $p = 0.001$)和 T2 母亲养育倦怠($\beta = 0.128$, $p = 0.007$)。为比较 T1 男生孤独感对 T2 父亲养育倦怠和 T2 母亲养育倦怠的预测作用是否存在差异,进行了 Wald 检验,结果显示差异未达到显著水平[χ^2(1) = 0.726, $p = 0.394$]。此外,还发现 T1 父亲养育倦怠对 T2 母亲养育倦怠的正向影响($\beta = 0.147$, $p = 0.008$)。没有发现其他显著的交叉滞后效应。因此,假设 2 得到支持,假设 1、3 和 4 没有得到支持。

女生模型的拟合结果如下:χ^2(1080) = 2653.180, $\chi^2/df = 2.457$; CFI = 0.932; TLI = 0.926; RMSEA = 0.054 (90% CI [0.051, 0.056]); SRMR = 0.068。整体来看,模型拟合良好。主观家庭社会经济地位对 T2 父亲养育倦怠的预测作用不显著($\beta = -0.052$, $p = 0.220$),对 T2 女生孤独感和 T2 母亲养育倦怠的负向预测作用显著($\beta = -0.127$, $p = 0.003$; $\beta = -0.137$, $p = 0.001$)。在控制了主观家庭社会经济地位后,父亲养育倦怠、母亲养育倦怠及女生孤独感在 T1 与 T2 之间均呈现出显著的自回归效应($\beta = 0.409$, $p < 0.001$; $\beta = 0.357$, $p < 0.001$; $\beta = 0.420$, $p < 0.001$)。T1 母亲养育倦怠能够显著正向预测 T2 女生孤独感($\beta = 0.185$, $p = 0.001$)。没有发现其他显著的交叉滞后效应。因此,假设 1 得到支持,假设 2 和 3 没有得到支持,假设 4 得到部分支持。

3.4 稳健性分析

青少年是否与父母同住是亲子互动过程中的重要家庭结构因素,可能通过改变亲子日常接触频率、情感交流机会以及父母对子女心理状态的觉察程度,进而关系到父母养育倦怠与早期青少年孤独感之间关联的方向和强度。为了检验本研究主要结果是否受到父母同住情况的影响,在仅保留青少年与父母双方共同居住的家庭后($N = 932$),重新进行了多组潜变量交叉滞后分析(同样控制了主观家庭社会经济地位)。结果显示,在男生中,T1 男生孤独

感能够显著正向预测 T2 父亲养育倦怠(β = 0.151, p = 0.002)和 T2 母亲养育倦怠(β = 0.139, p = 0.004),T1 父亲养育倦怠能够显著正向预测 T2 母亲养育倦怠(β = 0.153, p = 0.008)。在女生中,T1 母亲养育倦怠能够显著正向预测 T2 女生孤独感(β = 0.136, p = 0.022);T1 父亲养育倦怠能够显著正向预测 T2 母亲养育倦怠(β = 0.120, p = 0.038)。上述结果表明,男女生模型中亲子交叉滞后路径的显著性及方向均与主分析样本(N = 987)的结果一致,说明本研究的核心发现具有较好的稳健性。

4　讨论

尽管已有西方研究探讨父母养育倦怠与儿童青少年适应问题之间的双向关系,但在中国文化背景下,两者之间的关系模式仍不明确。同时,现有研究较少关注不同亲子性别组合的作用,未能充分揭示家庭内部的复杂互动关系。针对已有研究的不足,本研究深入考察中国父母养育倦怠和早期青少年孤独感之间的双向关系,并进一步考察不同亲子性别组合的作用。结果显示,在男生和女生中,本研究均没有发现父母养育倦怠和早期青少年孤独感之间存在双向关系;相反,本研究在男女生中发现了不同的单向影响模式,即男生孤独感影响父亲和母亲养育倦怠,表现为儿子驱动效应;在女生中则呈现出母亲影响女儿的特定路径,表现为母亲驱动效应。这些发现说明,父母养育倦怠与早期青少年孤独感之间的关系并非必然表现为普遍、对称的双向模式,这与西方家庭的研究结果不一致(Woine et al., 2026)。但这并不意味着父母养育倦怠与早期青少年孤独感之间缺乏关联,而是表明二者之间的关联可能具有较强的亲子性别组合特异性。不同于 Woine 等人(2026)将父母作为整体且未区分子女性别的做法,本研究同时区分父亲、母亲以及男生、女生,因而可能将潜在的双向关系进一步细化为更具体的性别化单向影响模式。也就是说,父母驱动效应和子女驱动效应并不一定在一种亲子性别组合中同时出现,而可能分别集中于不同的亲子性别组合。因此,本研究虽未发现预期的双向关系,但进一步凸显了亲子性别组合在识别家庭内部影响路径中的重要作用。

本研究发现,T1 时,父亲和母亲的养育倦怠水平在男生家庭与女生家庭之间均不存在显著差异,而 T2 时,男生的父亲和母亲均报告了更高水平的养育倦怠。这一从无性别差异到出现性别差异的变化初步提示,养育男孩的父母在后续阶段可能更容易出现养育倦怠水平升高的情况。进一步的交叉滞后分析显示,男生孤独感对父亲和母亲养育倦怠均有显著预测作用,这一发现在子女驱动效应的框架下同时支持了性别匹配效应与跨性别效应,但两种效应之间的差异未达到显著性水平。相比之下,女生孤独感对父亲和母亲养育倦怠均无显著预测作用,即未观察到女儿驱动效应。以往研究也发现了类似的性别差异,男生情绪问题可能引发父母的消极教养行为,而女生情绪问题的作用则不显著(Seely & Pössel, 2023; Xing & Wang, 2013)。性别社会化理论强调男生应表现出外向、坚强、独立(Fagot et al., 2000; Stockard, 2006),而抑郁、孤独等消极情绪被认为与传统男性气质不符(Mokhwelepa & Sumbane, 2025; Nordin et al., 2024)。当男生表现出孤独、退缩时,这种行为更容易被视为偏离传统男性气质,更难被社会接受(Rubin & Coplan, 2004),也更可能引发父母对其适应状况的担忧。虽然父亲对儿子的社会化负有更重要责任(Huston, 1983),但在“男孩偏好”以及“养儿防老”等传统观念的影响下(周钦, 袁燕, 2014),父亲和母亲可能都更加关注男生,并为其投入更多资源。在这种高期待和高投入的背景下,当男生面临适应问题时,可能同时引发父亲和母亲的焦虑,进而使父母表现出更高水平的养育倦怠。此外,情感脆弱性的污名化还会进一步削弱男生在经历消极情绪后的求助意愿(Mokhwelepa & Sumbane, 2025; Nordin et al., 2024)。在缺乏情绪表达和支持寻求的情况下,男生的孤独感可能更容易以外化行为的形式表现出来(Check et al., 1985; Moretta & Buodo, 2020; Wang et al., 2025; Wang & Zeng, 2024; 彭源 等, 2018)。例如,一项针对 10 岁至 13 岁韩国学生的纵向研究指出,男生内化问题会引发外化行为,在女生中则没有发现这一现象(Lee & Bukowski, 2012)。相比于孤独感等内化问题,外化行为更容易被父母识别,也更可能使父母双方在养育中都感受到更大的压力与情感耗竭。也就是说,男生孤独感还可能部分通过引发外化行为,进一步导致父母养育倦怠水平上升。

虽然在女生样本中未发现女儿驱动效应,但结果显示,母亲养育倦怠能够显著正向预测女儿孤独感,存在由母亲到女儿的单向影响路径。这一发现进一步强调,在具体亲子性别组合中,父母驱动效应和子女驱动效应可能分别以不同形式出现;同时,该

结果也从父母驱动效应的角度支持了亲子间的性别匹配效应,类似模式也得到既有研究支持。例如,母亲抑郁对子女抑郁的代际传递效应在女儿中更明显(Livings, 2021)。但与假设不一致的是,本研究并未在父亲养育倦怠与男生孤独感之间观察到相同的性别匹配效应。相反,Stone 等人(2016)的研究发现,父母较高的养育压力与男女生后续的内化问题均存在关联,并未表现出明显的性别差异。这可能是因为未区分父母角色,掩盖了不同亲子性别组合之间的异质性。本研究通过分别考察父亲和母亲的作用,识别出母亲影响女儿而非父亲影响儿子的特定作用路径,在一定程度上拓展并细化了既有研究的结论。具体而言,这一差异化结果可能恰与父母角色差异以及子女性别差异有关。一方面,“男主外,女主内”的传统性别角色分工在中国社会中仍占主流,母亲通常更多地参与育儿过程(贾云竹, 马冬玲, 2015; 王玮玲, 2016)。因此,当母亲出现养育倦怠并减少在育儿中的情感投入和互动时,子女更可能感知到这种变化,从而对其心理适应产生更为明显的影响;而与母亲相比,父亲参与日常照料的程度可能相对较低,因此其养育状态变化与子女心理适应之间的关联可能相对较弱。已有相关实证证据支持了这一解释,例如,王玮等人(2021)发现父亲养育倦怠对早期青少年孤独感的预测作用并不显著。平一凡等人(2022)发现母亲养育倦怠对早期青少年抑郁有显著正向预测作用,而父亲养育倦怠对早期青少年抑郁水平则没有显著预测作用。另一方面,女生也比男生表现出更高的拒绝敏感性(Maiolatesi et al., 2022)。因此,女生也更容易觉察到母亲在经历养育倦怠后情感支持的减少,从而对其社会情绪发展产生更强的影响。反之,男生对父亲养育状态变化的反应可能相对较弱。

本研究采用两波交叉滞后设计,在严格验证跨主体与跨时间联合测量不变性的基础上,系统考察了中国家庭中父母养育倦怠与早期青少年孤独感之间的双向关系以及亲子性别组合的作用。本研究同时纳入父亲、母亲和早期青少年的数据,有助于从多主体视角更全面地把握家庭系统内部的相互影响过程。在此基础上,本研究进一步揭示了不同亲子性别组合中的差异化作用模式,且研究结果具有较好的稳健性。这些发现补充了养育倦怠跨文化研究的实证证据,拓展了以往研究视角,并具有重要的理论与实践启示。在理论意义上,本研究将亲子性别组合与性别匹配效应纳入分析框架,进一步细化了发展相互作用模型中父母驱动效应和子女驱动效应的适用边界。研究发现,亲子间的相互影响具有特定的方向性与组合差异。具体而言,在父母驱动方向上,母亲养育倦怠仅显著预测女儿孤独感,契合了同性别亲子间心理联结更紧密的假设。在子女驱动方向上,男生孤独感同时预测父亲和母亲养育倦怠,但预测作用之间的差异不显著,说明在当前研究中子女驱动效应并未表现出同性别影响更强的特征。综上,亲子之间的相互影响并非仅由父母或子女单方面的性别决定,而是取决于具体的亲子性别组合。性别匹配效应也并非普遍存在,其适用性取决于驱动方向与亲子性别组合的共同作用。在实践意义上,本研究提示,相关家庭教育指导有必要进一步关注亲子性别组合所带来的差异化需求。对于养育男生的家庭,男生的孤独感是父母养育状态变化的显著预测因素之一。因此,家庭教育指导应帮助父母理性看待青春期男生的情绪情感变化,使其认识到孤独感可能是青春期早期同伴关系发展与自我同一性探索过程中较为常见的阶段性体验。这有助于父母以更具接纳性的态度回应男生的情感诉求,缓解因认知偏差或教养焦虑累积所致的养育倦怠。对于养育女生的家庭,本研究提示,母亲养育状态对女儿情绪适应具有更突出的作用。母女之间较强的情感联结,既可能增加母亲养育倦怠的后果向女儿传递的风险,也可能为发挥母亲积极养育的保护作用提供重要基础。母亲在日常照料中往往承担更多责任,因而更容易面临养育压力和情绪耗竭风险。因此,家庭教育指导应重视帮助母亲增强对自身养育情绪的觉察,认识到自身状态对女儿心理适应的潜在影响,并掌握有效的情绪调节与压力应对策略,以降低养育倦怠的后果向女儿传递的风险。在此基础上,还应积极引导母亲采用温暖、回应性的积极养育行为,从而促进女儿的情绪适应,并推动母女之间形成良性互动循环。

尽管具有上述优势与意义,本研究仍存在一些局限,有待未来研究进一步探讨:第一,本研究的样本仅包括初一年级入学后的早期青少年,未涵盖其他年龄段的儿童青少年,因此对研究结果的推广应当充分考虑当前样本年龄范围的限制。未来研究有必要纳入更广泛年龄段的样本,以进一步考察养育倦怠与青少年孤独感之间的双向关系在不同发展阶段的稳定性与差异性;第二,尽管本研究考虑了青少年是否与父母同住这一关键家庭结构因素的影响,但由于数据收集的限制,本研究未能考察子女是否

为独生子女这一潜在家庭结构因素的作用。一方面,同胞之间能够相互提供补偿性支持,缓解个体孤独感(Milevsky, 2005)。另一方面,更大规模的 PISA 调查发现,家庭规模(同胞数量)与青少年孤独感并无关联(Twenge et al., 2021);《中国国民心理健康发展报告(2021—2022)》(傅小兰 等, 2023)也显示,不同出生顺序的青少年在孤独感上的差异很微弱(η^2 = 0.001)。鉴于已有研究的不一致,未来研究可将子女是否为独生子女纳入分析,进一步考察父母养育倦怠与早期青少年孤独感之间的关系在独生子女家庭与非独生子女家庭中的差异;第三,本研究的相关结果仍有待在更多波次的纵向数据中进一步验证。尽管两波数据有助于考察变量之间的时间先后关系,但仍难以全面反映其动态变化过程。未来研究可以采用三波或更多时间点的数据,以更系统地刻画父母养育倦怠与青少年孤独感之间的动态发展过程;最后,本研究主要采用变量中心视角,可能掩盖了父母养育倦怠的群体异质性。未来研究可转向以个体为中心的视角(如潜在剖面分析),识别父母养育倦怠的不同亚型,并探讨其对不同性别青少年社会情绪发展的差异化影响。

5　结论

综上所述,本研究获得如下主要结论:(1)男生孤独感能够正向预测父亲养育倦怠和母亲养育倦怠;(2)母亲养育倦怠能够正向预测女生孤独感;(3)父亲和母亲养育倦怠与男女生孤独感之间均未发现双向关系。

参考文献:

Adler, N. E., Epel, E. S., Castellazzo, G., & Ickovics, J. R. (2000). Relationship of subjective and objective social status with psychological and physiological functioning: Preliminary data in healthy, White women. *Health Psychology*, *19*(6), 586 – 592.

Asher, S. R., Hymel, S., & Renshaw, P. D. (1984). Loneliness in children. *Child Development*, *55*(4), 1456 – 1464.

Ayhan, A. B., & Beyazit, U. (2021). The associations between loneliness and self – esteem in children and neglectful behaviors of their parents. *Child Indicators Research*, *14*(5), 1863 – 1879.

Bandura, A. (1977). *Social learning theory*. Prentice – Hall.

Bian, H., Bian, Y., & Liang, L. (2025). The impact of parental burnout on family cohesion: The mediating roles of parent – child relationships and parental differences. *Current Psychology*, *44*(10), 9435 – 9445.

Bian, H., Pu, Q., He, Y., Zhang, Y., & Chen, F. (2026). How does mothers' parental burnout longitudinally predict adolescent bullying perpetration and victimization? The mediating role of psychological control rather than autonomy support. *Journal of Youth and Adolescence*. *55*(8), 2096 – 2112.

Blanchard, L. T., Gurka, M. J., & Blackman, J. A. (2006). Emotional, developmental, and behavioral health of American children and their families: A report from the 2003 National Survey of Children's Health. *Pediatrics*, *117*(6), e1202 – e1212.

Brinker, V., Dewald – Kaufmann, J., Padberg, F., & Reinhard, M. A. (2023). Aggressive intentions after social exclusion and their association with loneliness. *European Archives of Psychiatry and Clinical Neuroscience*, *273*(5), 1023 – 1028.

Check, J. V. P., Perlman, D., & Malamuth, N. M. (1985). Loneliness and aggressive behaviour. *Journal of Social and Personal Relationships*, *2*(3), 243 – 252.

Chen, B. – B., Qu, Y., Yang, B., & Chen, X. (2022). Chinese mothers' parental burnout and adolescents' internalizing and externalizing problems: The mediating role of maternal hostility. *Developmental Psychology*, *58*(4), 768 – 777.

Chen, F. (2007). Sensitivity of goodness of fit indexes to lack of measurement invariance. *Structural Equation Modeling: A Multidisciplinary Journal*, *14*(3), 464 – 504.

Cheng, H., Wang, W., Wang, S., Li, Y., Liu, X., & Li, Y. (2020). Validation of a Chinese version of the Parental Burnout Assessment. *Frontiers in Psychology*, *11*, Article 321. https://doi.org/10.3389/fpsyg.2020.00321

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Routledge.

Crouter, A. C., Manke, B. A., & McHale, S. M. (1995). The family context of gender intensification in early adolescence. *Child Development*, *66*(2), 317 – 329.

de Santis, L., Noronha, A. P. P., Mikolajczak, M., & Roskam, I. (2025). A step forward in understanding gender differences in parental burnout: Parental involvement and emotion regulation. *Journal of Child and Family Studies*, *34*(1), 95 – 108.

Fagot, B. I., Rodgers, C. S., & Leinbach, M. D. (2000). Theories of gender socialization. In T. Eckes & H. M. Trautner (Eds.), *The developmental social psychology of gender* (pp. 65 – 89). Lawrence Erlbaum Associates Publishers.

González – Abaurrea, E., González – Cabrera, J., Ortega – Barón, J., Escortell, R., & Machimbarrena, J. M. (2026). Bidirectional relationships between peer victimization, loneliness and solitude in adolescence: A systematic review. *Adolescent Research Review*, *11*(1), 195 – 247.

Grygiel, P., Dolata, R., Humenny, G., & Muszyński, M. (2024). Depressive symptoms and loneliness among early adolescents: A psychometric network analysis approach. *Journal of Child Psychology and Psychiatry*, *65*(2), 199 – 214.

Hong, V. T. P., An, N. H., Thao, T. T. P., Thao, L. N., & Thanh, N. M. (2022). Behavior problems reduce academic outcomes among primary students: A moderated mediation of parental burnout and parents' self – compassion. *New Directions for Child and Adolescent Development*, (185 – 186), 27 – 42.

Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal*, *6*(1), 1 – 55.

Hughes, E. K., & Gullone, E. (2010). Reciprocal relationships between parent and adolescent internalizing symptoms. *Journal of Family Psychology*, *24*(2), 115 – 124.

Huston, A. C. (1983). Sex – typing. In E. M. Hetherington (Ed.), *Handbook of child psychology: Vol. 4. Socialization, personality, and social development* (pp. 387 – 467). Wiley.

Laursen, B., & Collins, W. A. (2004). Parent – child communication during adolescence. In A. L. Vangelisti (Ed.), *Handbook of family communication* (pp. 333 – 348). Lawrence Erlbaum Associates Publishers.

Lee, E. J., & Bukowski, W. M. (2012). Co – development of internalizing and externalizing problem behaviors: Causal direction and common vulnerability. *Journal of Adolescence*, *35*(3), 713 – 729.

Li, X., & Meier, J. (2017). Father love and mother love: Contributions of parental acceptance to children's psychological adjustment. *Journal of Family Theory & Review*, *9*(4), 459 – 490.

Livings, M. S. (2021). The gendered relationship between maternal depression and adolescent internalizing symptoms. *Social Science & Medicine*, *291*, Article 114464. https://doi.org/10.1016/j.socscimed.2021.114464

Lodder, G. M. A., Goossens, L., Scholte, R. H. J., Engels, R. C. M. E., & Verhagen, M. (2016). Adolescent loneliness and social skills: Agreement and discrepancies between self – , meta – , and peer – evaluations. *Journal of Youth and Adolescence*, *45*(12), 2406 – 2416.

Maiolatesi, A. J., Clark, K. A., & Pachankis, J. E. (2022). Rejection sensitivity across sex, sexual orientation, and age: Measurement invariance and latent mean differences. *Psychological Assessment*, *34*(5), 431 – 442.

Matthews, T., Caspi, A., Danese, A., Fisher, H. L., Moffitt, T. E., & Arseneault, L. (2022). A longitudinal twin study of victimization and loneliness from childhood to young adulthood. *Development and Psychopathology*, *34*(1), 367 – 377.

McSherry, D., Fargas Malet, M., & Weatherall, K. (2019). The Strengths and Difficulties Questionnaire (SDQ): A proxy measure of parenting stress. *The British Journal of Social Work*, *49*(1), 96 – 115.

Mikolajczak, M., Aunola, K., Sorkkila, M., & Roskam, I. (2023). 15 years of parental burnout research: Systematic review and agenda. *Current Directions in Psychological Science*, *32*(4), 276 – 283.

Mikolajczak, M., Brianda, M. E., Avalosse, H., & Roskam, I. (2018). Consequences of parental burnout: Its specific effect on child neglect and violence. *Child Abuse & Neglect*, *80*, 134 – 145.

Mikolajczak, M., Gross, J. J., & Roskam, I. (2019). Parental burnout: What is it, and why does it matter? *Clinical Psychological Science*, *7*(6), 1319 – 1329.

Milevsky, A. (2005). Compensatory patterns of sibling support in emerging adulthood: Variations in loneliness, self – esteem, depression and life satisfaction. *Journal of Social and Personal Relationships*, *22*(6), 743 – 755.

Minh, T. N., Quang, A. M. T., & Ha, T. A. (2022). The influence of children's behavior problems on parents' psychological well – being: A serial mediation model of parental psychological control and parental burnout. *Children and Youth Services Review*, *134*, Article 106366. https://doi.org/10.1016/j.childyouth.2022.106366

Mokhwelepa, L. W., & Sumbane, G. O. (2025). Men's mental health matters: The impact of traditional masculinity norms on men's willingness to seek mental health support; A systematic review of literature. *American Journal of Men's Health*, *19*(3), Article 15579883251321670. https://doi.org/10.1177/15579883251321670

Moretta, T., & Buodo, G. (2020). Problematic internet use and loneliness: How complex is the relationship? A short literature review. *Current Addiction Reports*, *7*(2), 125 – 136.

Neece, C. L., Green, S. A., & Baker, B. L. (2012). Parenting stress and child behavior problems: A transactional relationship across time. *American Journal on Intellectual and Developmental Disabilities*, *117*(1), 48 – 66.

Nordin, T., Degerstedt, F., & Granholm Valmari, E. (2024). A scoping review of masculinity norms and their interplay with loneliness and social connectedness among men in Western societies. *American Journal of Men's Health*, *18*(6), Article 15579883241304585. https://doi.org/10.1177/15579883241304585

Qualter, P., Vanhalst, J., Harris, R., Van Roekel, E., Lodder, G., Bangee, M., … Verhagen, M. (2015). Loneliness across the life span. *Perspectives on Psychological Science*, *10*(2), 250 – 264.

Ren, P., Wang, Y., Liang, Y., Li, S., & Wang, Q. (2023). Bidirectional relationship between bullying victimization and functions of aggression in adolescents: The mediating effect of teacher justice. *Journal of Adolescence*, *95*(6), 1245 – 1257.

Ren, X., Cai, Y., Wang, J., & Chen, O. (2024). A systematic review of parental burnout and related factors among parents. *BMC Public Health*, *24*(1), Article 376. https://doi.org/10.1186/s12889 – 024 – 17829 – y

Roskam, I., Brianda, M. – E., & Mikolajczak, M. (2018). A step forward in the conceptualization and measurement of parental burnout: The Parental Burnout Assessment (PBA). *Frontiers in Psychology*, *9*, Article 758. https://doi.org/10.3389/fpsyg.2018.00758

Roskam, I., Raes, M. – E., & Mikolajczak, M. (2017). Exhausted parents: Development and preliminary validation of the Parental Burnout Inventory. *Frontiers in Psychology*, *8*, Article 163. https://doi.org/10.3389/fpsyg.2017.00163

Rubin, K. H., & Coplan, R. J. (2004). Paying attention to and not neglecting social withdrawal and social isolation. *Merrill – Palmer Quarterly*, *50*(4), 506 – 534.

Rueter, M. A., & Conger, R. D. (1998). Reciprocal influences between parenting and adolescent problem – solving behavior. *Developmental Psychology*, *34*(6), 1470 – 1482.

Salimi, A., & Bozorgpour, F. (2012). Percieved social support and social – emotional loneliness. *Procedia – Social and Behavioral Sciences*, *69*, 2009 – 2013.

Sameroff, A. J. (1975). Transactional models in early social relations.

Human Development, *18*(1 – 2), 65 – 79.

Sameroff, A. J., & MacKenzie, M. J. (2003). Research strategies for capturing transactional models of development: The limits of the possible. *Development and Psychopathology*, *15*(3), 613 – 640.

Seely, H. D., & Pössel, P. (2023). Parenting behavior and adolescent affect: Bidirectionality and the role of gender. *Journal of Family Issues*, *44*(5), 1321 – 1342.

Solholm, R., Bjørknes, R., & Jakobsen, R. (2019). Change in mothers' perception of well – being and distress following treatment of child conduct problems. *Journal of Child and Family Studies*, *28*(4), 1059 – 1070.

Sorhagen, N. S., & Wurster, T. J. (2017). Income within context: Relative income matters for adolescent social satisfaction and mental health. *Journal of Child Psychology and Psychiatry*, *58*(6), 736 – 743.

Stockard, J. (2006). Gender socialization. In J. S. Chafetz (Ed.), *Handbook of the sociology of gender* (pp. 215 – 227). Springer.

Stone, L. L., Mares, S. H. W., Otten, R., Engels, R. C. M. E., & Janssens, J. M. A. M. (2016). The co – development of parenting stress and childhood internalizing and externalizing problems. *Journal of Psychopathology and Behavioral Assessment*, *38*(1), 76 – 86.

Suárez – Relinque, C., Del Moral, G., León – Moreno, C., & Callejas – Jerónimo, J. E. (2023). Emotional loneliness, suicidal ideation, and alexithymia in adolescents who commit child – to – parent violence. *Journal of Interpersonal Violence*, *38*(3 – 4), 4007 – 4033.

Sun, Z., Ding, W., Li, J., Jiang, M., Li, W., & Xie, R. (2024). Why am I always lonely? The lasting impact of childhood harsh parental discipline on adolescents' loneliness. *Family Relations*, *73*(3), 1933 – 1948.

Twenge, J. M., Haidt, J., Blake, A. B., McAllister, C., Lemon, H., & Le Roy, A. (2021). Worldwide increases in adolescent loneliness. *Journal of Adolescence*, *93*(1), 257 – 269.

Wang, J., Yang, Y., Chen, Y., Lin, H., Wang, T., Wang, Z., … Fu, C. (2025). Loneliness, internalizing and externalizing problems, and suicidal ideation among Chinese adolescents: A longitudinal mediation analysis. *Journal of Adolescent Health*, *76*(1), 96 – 104.

Wang, Y., & Zeng, Y. (2024). Relationship between loneliness and internet addiction: A meta – analysis. *BMC Public Health*, *24*(1), Article 858. https://doi.org/10.1186/s12889 – 024 – 18366 – 4

Wheaton, B., Muthén, B., Alwin, D. F., & Summers, G. F. (1977). Assessing reliability and stability in panel models. *Sociological Methodology*, *8*, 84 – 136.

Woine, A., Dardier, V., Mikolajczak, M., & Roskam, I. (2026). Bidirectional and transactional effects between parental burnout and child internalizing and externalizing behaviors. *Journal of Family Psychology*, *40*(3), 447 – 457.

Xing, X., & Wang, M. (2013). Sex differences in the reciprocal relationships between mild and severe corporal punishment and children's internalizing problem behavior in a Chinese sample. *Journal of Applied Developmental Psychology*, *34*(1), 9 – 16.

Xu, J., Ni, S., Ran, M., & Zhang, C. (2017). The relationship between parenting styles and adolescents' social anxiety in migrant families: A study in Guangdong, China. *Frontiers in Psychology*, *8*, Article 626. https://doi.org/10.3389/fpsyg.2017.00626

Yang, B., Chen, B. – B., Qu, Y., & Zhu, Y. (2021). Impacts of parental burnout on Chinese youth's mental health: The role of parents' autonomy support and emotion regulation. *Journal of Youth and Adolescence*, *50*(8), 1679 – 1692.

Yang, B., Zhou, Z., Qu, Y., & Chen, B. – B. (2025). Is there a vicious cycle between parental burnout and parent – adolescent conflict? A three – wave within – family analytic approach. *Family Process*, *64*(1), Article e70015. https://doi.org/10.1111/famp.70015

Yang, M., Wei, W., Ren, L., Pu, Z., Zhang, Y., Li, Y., … Wu, S. (2023). How loneliness linked to anxiety and depression: A network analysis based on Chinese university students. *BMC Public Health*, *23*(1), Article 2499. https://doi.org/10.1186/s12889 – 023 – 17435 – 4

Zheng, M., Gao, Y., Li, J., & Liu, X. (2025). Longitudinal relationship between aggressive behavior and non – suicidal self – injury among adolescent boys and girls: The mediating role of peer victimization. *Journal of Youth and Adolescence*, *54*(6), 1506 – 1519.

Zou, S., Wu, X., & Li, X. (2020). Coparenting behavior, parent – adolescent attachment, and peer attachment: An examination of gender differences. *Journal of Youth and Adolescence*, *49*(1), 178 – 191.

白春玉, 张迪, 陈容, 顾国家. (2011). 父母教养方式对中学生孤独感的影响. *中国学校卫生*, *32*(4), 473 – 474.

边昊天, 庄瑞雪, 王嘉宝, 边玉芳, 梁丽婵. (2026). 养育倦怠与青少年网络游戏成瘾:基本心理需要满足、意志控制及性别的作用. *心理发展与教育*, *42*(1), 110 – 120.

董奇, 林崇德. (编). (2011). *中国儿童青少年心理发育标准化测验简介*. 中国:科学出版社.

傅小兰, 张侃, 陈雪峰, 陈祉妍. (编). (2023). *中国国民心理健康发展报告(2021——2022)*. 中国:社会科学文献出版社.

耿梦龙, 陶芳标. (2018). 儿童内外化行为共病现象及对未来发展的影响. *中国学校卫生*, *39*(4), 637 – 640.

贾云竹, 马冬玲. (2015). 性别观念变迁的多视角考量:以"男主外,女主内"为例. *妇女研究论丛*, (3), 29 – 36.

李森, 张双喜, 俞国良. (2024). 父母倦怠对中小学生心理健康的影响:证据与教育对策. *教育科学研究*, (10), 12 – 19.

彭源, 朱蕾, 王振宏. (2018). 父母情绪表达与青少年问题行为:亲子依恋、孤独感的多重中介效应. *心理发展与教育*, *34*(4), 504 – 512.

平一凡, 刘淘淘, 王玮, 李永鑫. (2022). 父母抑郁在养育倦怠与子女抑郁间的中介作用. *中国临床心理学杂志*, *30*(4), 964 – 967.

秦瑶, 彭运石. (2024). 父母教养方式对初中生社交焦虑的影响:同伴接纳和反刍思维的链式中介作用. *心理发展与教育*, *40*(1), 103 – 113.

宋广文, 郝丙辉. (2005). 学生社交回避行为、苦恼体验状况及其与父母教养方式关系的研究. *中国健康心理学杂志*, *13*(2), 133 – 135.

王凌飞, 张航, 边玉芳. (2026). 个体水平及群体水平的同伴支持与青少年内化问题的关系:基于社会网络分析方法. *心理科学*,

49(1), 105 – 120.

王明忠, 范翠英, 周然. (2013). 父母情感温暖与青少年孤独感的关系:序列中介效应分析. *心理与行为研究*, *11*(5), 654 – 659.

王玮, 王胜男, 程华斌, 王云鹤, 李永鑫. (2021). 父亲养育倦怠在养育压力与青少年心理健康之间的中介作用. *中国临床心理学杂志*, *29*(4), 858 – 861.

王玮玲. (2016). 基于性别的家庭内部分工研究. *重庆大学学报(社会科学版)*, *22*(5), 135 – 143.

周钦, 袁燕. (2014). 家庭基础教育投入决策"男孩偏好"的理论与实证研究. *人口学刊*, *36*(3), 14 – 24.

邹泓. (1999). 中学生的社会支持系统与同伴关系. *北京师范大学学报(社会科学版)*, (1), 34 – 42.

Reciprocal Relationships between Parental Burnout and Early Adolescent Loneliness: An Analysis of Parent-adolescent Gender Dyads

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Abstract: To examine the reciprocal relationships between parental burnout and early adolescent loneliness and the role of parent-adolescent gender dyads, this study employed a two-wave cross-lagged design, with data collected when adolescents were in Grade 7 and again in Grade 8. Fathers' and mothers' parental burnout was assessed using the Parental Burnout Assessment, and early adolescents' loneliness was assessed using the Children and Adolescents Loneliness Scale. Data from 987 families were included in the final analysis. In Grade 7, early adolescents had a mean age of 13.54 ± 0.34 years, fathers had a mean age of 43.34 ± 5.59 years, and mothers had a mean age of 41.31 ± 4.63 years. After controlling for subjective family socioeconomic status, cross-lagged panel analyses revealed that, among boys, loneliness positively predicted both fathers' and mothers' parental burnout. Among girls, mothers' parental burnout positively predicted girls' loneliness. Furthermore, no reciprocal relationships were found between fathers' or mothers' parental burnout and boys' or girls' loneliness. These findings indicate that the associations between parental burnout and early adolescent loneliness are characterized by unidirectional predictive patterns that vary across parent-adolescent gender dyads.

Key words: parental burnout; loneliness; reciprocal relationships; parent-adolescent gender dyads; early adolescents